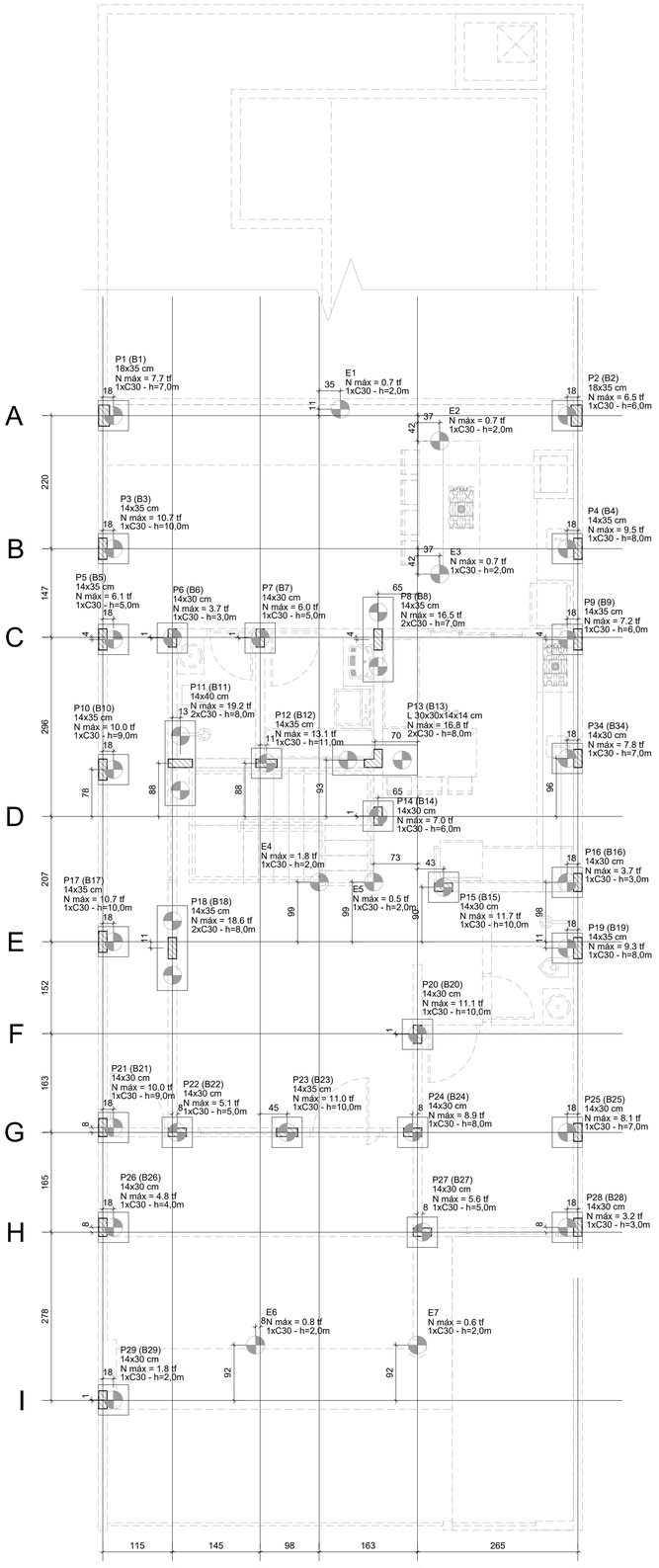
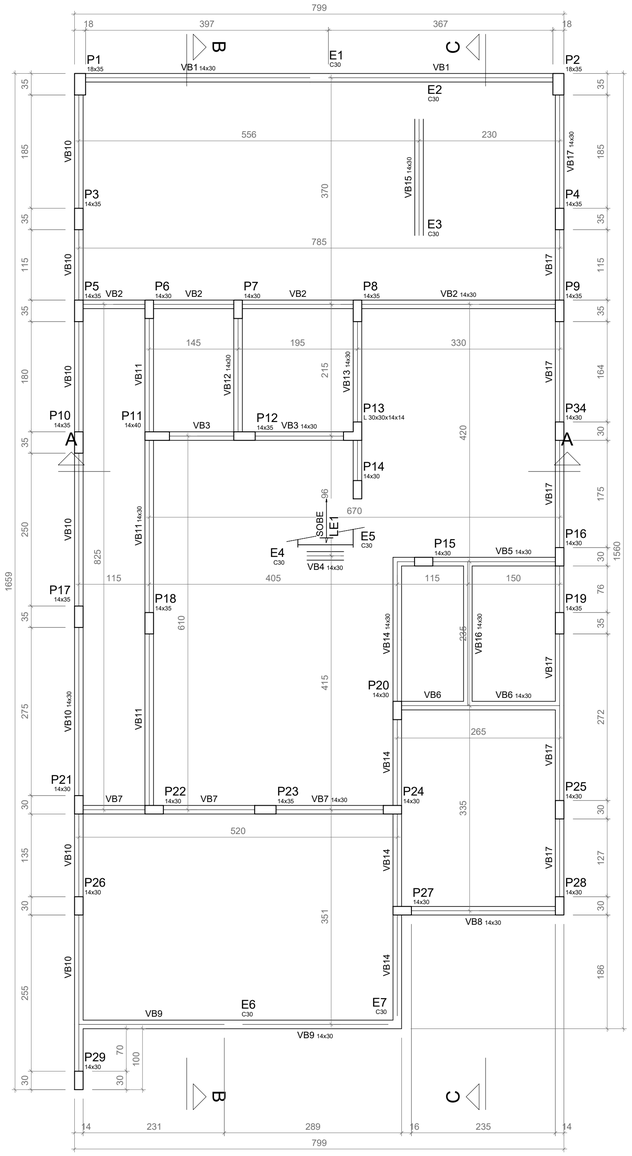


Planta de locação

escala 1:50

Vigas			
Nome	Seção (cm)	Elevação (cm)	Nível (cm)
VB1	14x30	0	35
VB2	14x30	0	35
VB3	14x30	0	35
VB4	14x30	0	35
VB5	14x30	0	35
VB6	14x30	0	35
VB7	14x30	0	35
VB8	14x30	0	35
VB9	14x30	0	35
VB10	14x30	0	35
VB11	14x30	0	35
VB12	14x30	0	35
VB13	14x30	0	35
VB14	14x30	0	35
VB15	14x30	0	35
VB16	14x30	0	35
VB17	14x30	0	35

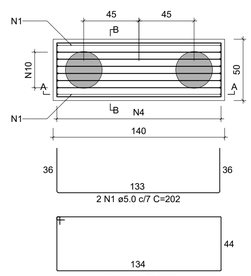


Forma do pavimento Baldrame (Nível 35)

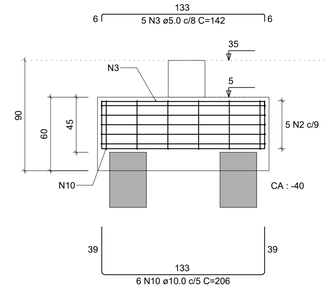
escala 1:50

B11=B13
2xC30

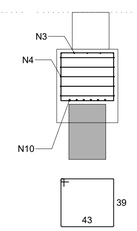
PLANTA
ESC 1:25



CORTE A-A
ESC 1:25

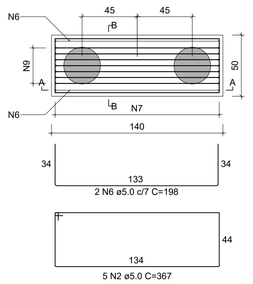


CORTE B-B
ESC 1:25

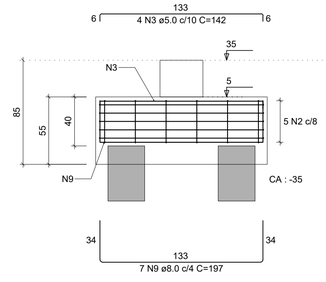


B8=B18
2xC30

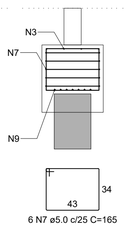
PLANTA
ESC 1:25



CORTE A-A
ESC 1:25

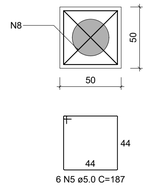


CORTE B-B
ESC 1:25

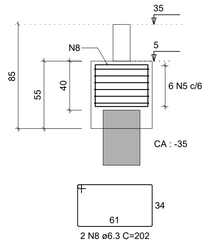


B1=B2=B3=B4=B5=B6=B7=B9=B10=B12=B14=B15
=B16=B17=B19=B20=B21=B22=B23=B24=B25
=B26=B27=B28=B29=B34
1xC30

PLANTA
ESC 1:25



CORTE
ESC 1:25



RELAÇÃO DO AÇO

2xB13 26xB15 2xB18

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	4	202	808
	2	5.0	20	367	7340
	3	5.0	18	142	2556
	4	5.0	12	175	2100
	5	5.0	156	187	29172
	6	5.0	4	198	792
	7	5.0	12	165	1980
CA50	8	6.3	52	202	10504
	9	8.0	14	197	2758
	10	10.0	12	206	2472

RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	6.3	105	28.3
	8.0	27.6	12
	10.0	24.7	16.8
CA60	5.0	447.5	75.9
PESO TOTAL (kg)			
CA50	57		
CA60	75.9		

Volume de concreto (C-25) = 4.82 m³
Área de forma = 37.34 m²

OBSERVAÇÕES		TABELA DE EQUIV. AÇOS	
SOBRECARGA NAS LAJES		mm	Pol.
200Kgf/m2	SEM ALVENARIA	5,0	3/16"
		6,3	1/4"
		8,0	5/16"
		10	3/8"
300Kgf/m2	COM ALVENARIA	12,5	1/2"
		16	5/8"
		20	3/4"

LEGENDA PILARES E VIGAS:

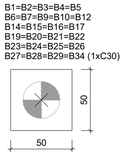
- Pilar que nasce
- Pilar com mudança de seção
- Pilar que continua
- Viga
- Pilar que morre
- Viga chata ou invertida
- Pilarete (PIL)
- Lxx/ hyy — Numero da laje, sentido da viguta e altura da laje
- Alvenaria sobre laje

ESPECIFICAÇÕES:
- CONFRONTAR PROJETO ESTRUTURAL COM PROJETO ARQUITETÔNICO E LOCAÇÃO EM OBRA
- EM LOCAIS COM ATERRO, ACRESCENTAR A ALTURA DO ATERRO NA PROFUNDIDADE DAS FUNDAÇÕES E EM SUAS ARMADURAS.
- ANTES DO INÍCIO DA CONCRETAGEM, AS FORMAS DEVERÃO ESTAR LIMPAS E ESTANQUES DE MODO A EVITAR EVENTUAIS FUGAS DE PASTA.
- AS FORMA DEVERÃO SER MOLHADAS ATÉ A SATURAÇÃO A FIM DE EVITAR A ABSORÇÃO DA ÁGUA DE AMASSAMENTO DO CONCRETO
- AS BARRAS DE AÇO NÃO DEVEM APRESENTAR FERRUGEM, MANCHAS DE ÓLEO OU QUAISQUER OUTRAS SUBSTÂNCIAS QUE IMPEÇAM UMA PERFEITA ADERÊNCIA AO CONCRETO
- AS ARMADURAS NÃO DEVERÃO FICAR EM CONTATO DIRETO COM AS FORMAS, OBEDECENDO PARA ISSO OS COBRIMENTOS MÍNIMOS
- O AÇO UTILIZADO NA ESTRUTURA DEVE SER CA 50A E CA 60.
- O ADENSAMENTO É OBRIGATÓRIO E DEVERÁ SER CUIDADOSO, OCUPANDO TODOS OS RECANTOS DA FORMA, EVITANDO A VIBRAÇÃO DAS ARMADURAS QUE PODE PROVOCAR VAZIOS AO REDOR DAS ARMADURAS, DIFICULTANDO A ADERÊNCIA DO CONCRETO
- O PROCESSO DE CURA DO CONCRETO SERÁ NO MÍNIMO DE SETE DIAS

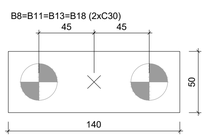
- PRAZOS RECOMENDADOS PARA DESFORMA:
  FACES LATERAIS: 03 DIAS
  FACES INFERIORES: 14 DIAS
  FACES INFERIORES SEM PONTALETES: 21 DIAS
- A RETIRADA DOS ESCORAMENTOS DOS TETOS DEVERÁ SER FEITA DE MANEIRA CONVENIENTE E PROGRESSIVA, PARTICULARMENTE PARA AS PEÇAS EM BALANÇO, O QUE IMPEDIRÁ O APARECIMENTO DE FISSURAS EM DECORRÊNCIA DE CARGAS DIFERENCIAIS
- A LAJE PRÉ-MOLDADA É DE RESPONSABILIDADE DA EMPRESA FORNECEDORA DE LAJES
- MEDIDAS APRESENTADAS NO PROJETO EM CENTIMETROS (CM)
- QUALQUER DÚVIDA CONSULTAR O ENGENHEIRO CALCULISTA

CONCRETO: FCK=250Kgf/cm²

B1=B2=B3=B4=B5
B6=B7=B9=B10=B12
B14=B15=B16=B17
B19=B20=B21=B22
B23=B24=B25=B26
B27=B28=B29=B34 (1xC30)



B8=B11=B13=B18 (2xC30)

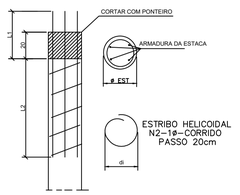


Legenda dos blocos

escala 1:25

DETALHE PARA ARMADURA DAS ESTACAS

		ARMADURA LONGITUDINAL			ARMADURA TRANSVERSAL		
Ø EST.	QUANT.	ARMADURA	L1(cm)	L2(cm)	ESTRIBO	FORM.	di(cm)
30	33	4N1 Ø8 C=240	50	190	1N2 Ø5 c/20 C=150	helic.	24
30	8	4N1 Ø8 C=150	50	100	1N2 Ø5 c/20 C=150	helic.	24
TOTAL	41						



RESUMO DO AÇO - ESTACAS

AÇO	DIAM	C. TOTAL (m)	PESO + 10% (kg)
CA60	5	684,0	115,87
CA50	8	372,0	161,63
PESO TOTAL (kg)		277,50	

VOL. CONCRETO	
C15	19,67 m³

IMPORTANTE
- OS COMPRIMENTOS DAS ARMADURAS E PROFUNDIDADES DAS ESTACAS SÃO REFERENTES AO SOLO NATURAL.
- CASO SEJA NECESSÁRIO ATERRO, PROLONGAR A ARMADURA E PERFURAÇÃO NA MESMA PROFUNDIDADE ATERRADA.

PROJETO ESTRUTURAL

OBRA: CONSTRUÇÃO DE UMA EDIFICAÇÃO HABITACIONAL COM 2 PAVIMENTOS EM ALVENARIA

PROPR.:

PROJETO:

Geminada.com

FOLHA Nº: 01/09

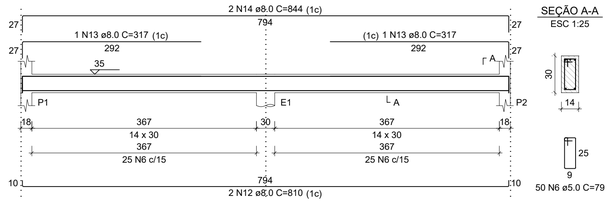
REVISÃO: R-01

DATA: 03/2019

REF.:
LOCAÇÃO FUNDAÇÃO
DET. ESTACAS e BLOCOS DE FUNDAÇÃO
FORMA - PAV. TERREO

CONCRETO: 25 MPa

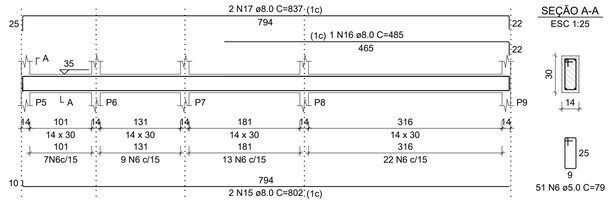
VB1
ESC 1:50



SEÇÃO A-A
ESC 1:25

50 N6 ø5.0 C=79

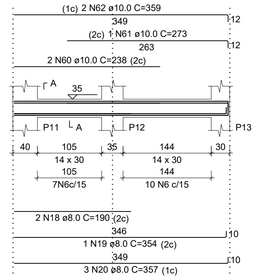
VB2
ESC 1:50



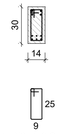
SEÇÃO A-A
ESC 1:25

51 N6 ø5.0 C=79

VB3
ESC 1:50

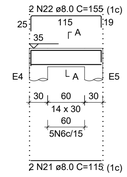


SEÇÃO A-A
ESC 1:25

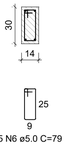


17 N6 ø5.0 C=79

VB4
ESC 1:50

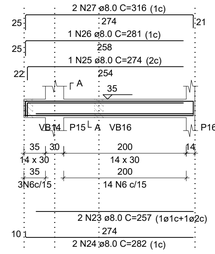


SEÇÃO A-A
ESC 1:25

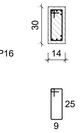


5 N6 ø5.0 C=79

VB5
ESC 1:50



SEÇÃO A-A
ESC 1:25



17 N6 ø5.0 C=79

RELAÇÃO DO AÇO

2xP1	P3	3xP4
2xP5	6xP6	3xP10
P11	2xP12	P13
5xP14	P15	P22
P28	P29	VB1
VB2	VB3	VB4
VB5	VB6	VB7
VB8	VB9	VB10
VB11	VB12	VB13
VB14	VB15	VB16
VB17		

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	10	28	280
	2	5.0	10	97	970
	3	5.0	14	24	336
	4	5.0	77	89	6853
	5	5.0	56	24	1344
	6	5.0	676	79	53404
	7	5.0	22	24	528
	8	5.0	5	99	495
	9	5.0	6	65	390
	10	5.0	14	55	770
	11	5.0	1	100	100
CA50	12	8.0	2	810	1620
	13	8.0	2	317	634
	14	8.0	2	844	1688
	15	8.0	2	802	1604
	16	8.0	1	485	485
	17	8.0	2	837	1674
	18	8.0	2	190	380
	19	8.0	1	354	354
	20	8.0	3	357	1071
	21	8.0	2	115	230
	22	8.0	2	155	310
	23	8.0	2	257	514
	24	8.0	4	282	1128
	25	8.0	1	274	274
	26	8.0	1	281	281
	27	8.0	2	316	632
	28	8.0	3	290	870
	29	8.0	2	315	630
	30	8.0	3	537	1611
	31	8.0	1	550	550
	32	8.0	3	553	1659
	33	8.0	3	299	897
	34	8.0	2	545	1090
	35	8.0	2	571	1142
	36	8.0	2	1084	2168
	37	8.0	2	630	1260
	38	8.0	1	320	320
	39	8.0	2	1198	2396
	40	8.0	2	521	1042
	41	8.0	2	834	1668
	42	8.0	2	851	1702
	43	8.0	2	224	448
	44	8.0	2	241	482
	45	8.0	4	320	1280
	46	8.0	2	773	1546
	47	8.0	2	807	1614
	48	8.0	2	245	490
	49	8.0	2	279	558
	50	8.0	2	244	488
	51	8.0	2	278	556
	52	8.0	1	365	365
	53	8.0	2	1164	2328
	54	8.0	2	265	530
	55	8.0	1	545	545
	56	8.0	2	1200	2400
	57	8.0	2	230	460
	58	10.0	112	VAR	VAR
	59	10.0	4	VAR	VAR
	60	10.0	2	238	476
	61	10.0	1	273	273
	62	10.0	2	359	718
	63	12.5	16	VAR	VAR
	64	12.5	8	VAR	VAR
	65	16.0	24	VAR	VAR
	66	16.0	6	148	888

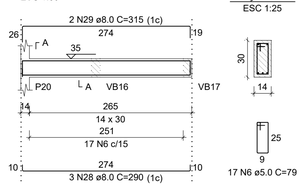
RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	8.0	459.7	199.5
	10.0	155.7	105.6
	12.5	31.8	33.7
	16.0	43.2	75
CA60	5.0	654.7	111

PESO TOTAL (kg)	
CA50	413.9
CA60	111

Volume de concreto (C-25) = 5.07 m³
Área de forma = 91.67 m²

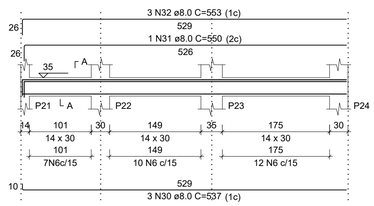
VB6
ESC 1:50



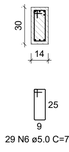
SEÇÃO A-A
ESC 1:25

17 N6 ø5.0 C=79

VB7
ESC 1:50

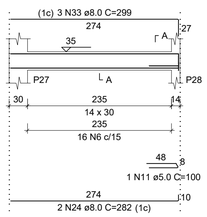


SEÇÃO A-A
ESC 1:25

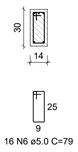


29 N6 ø5.0 C=79

VB8
ESC 1:50

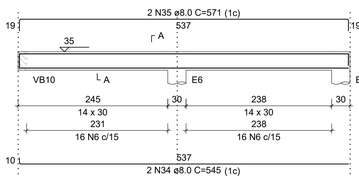


SEÇÃO A-A
ESC 1:25

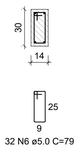


16 N6 ø5.0 C=79

VB9
ESC 1:50

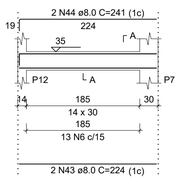


SEÇÃO A-A
ESC 1:25

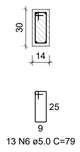


32 N6 ø5.0 C=79

VB12
ESC 1:50

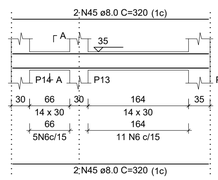


SEÇÃO A-A
ESC 1:25

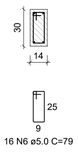


13 N6 ø5.0 C=79

VB13
ESC 1:50

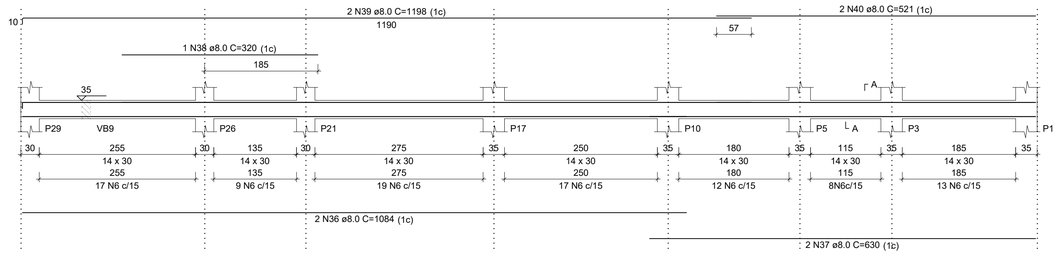


SEÇÃO A-A
ESC 1:25

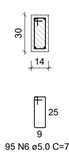


16 N6 ø5.0 C=79

VB10
ESC 1:50

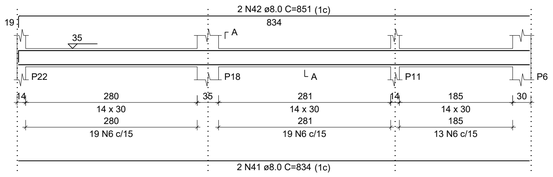


SEÇÃO A-A
ESC 1:25

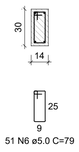


95 N6 ø5.0 C=79

VB11
ESC 1:50

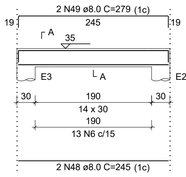


SEÇÃO A-A
ESC 1:25

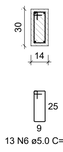


51 N6 ø5.0 C=79

VB15
ESC 1:50

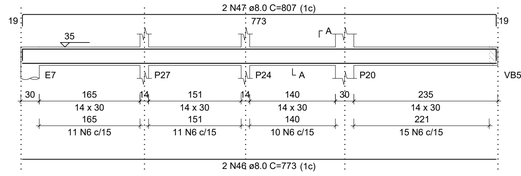


SEÇÃO A-A
ESC 1:25



13 N6 ø5.0 C=79

VB14
ESC 1:50

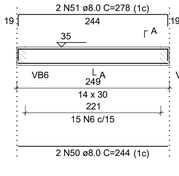


SEÇÃO A-A
ESC 1:25

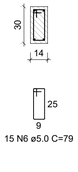


47 N6 ø5.0 C=79

VB16
ESC 1:50

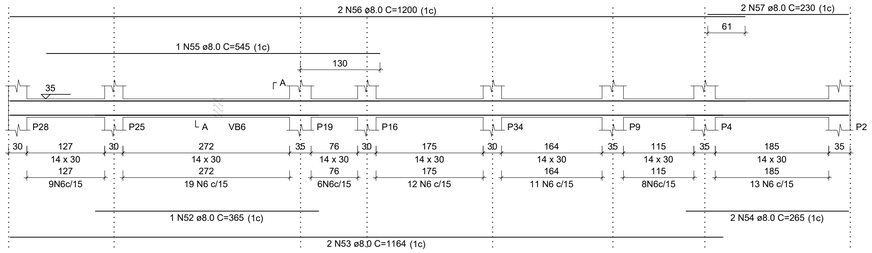


SEÇÃO A-A
ESC 1:25

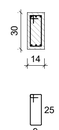


15 N6 ø5.0 C=79

VB17
ESC 1:50

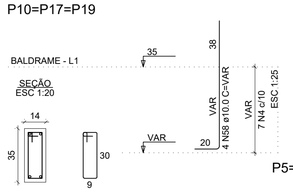


SEÇÃO A-A
ESC 1:25

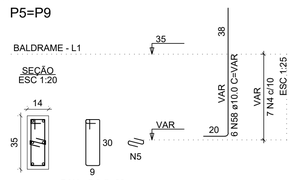


78 N6 ø5.0 C=79

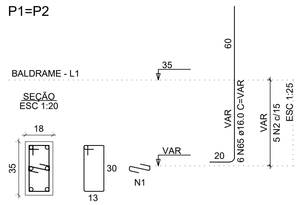
P10=P17=P19



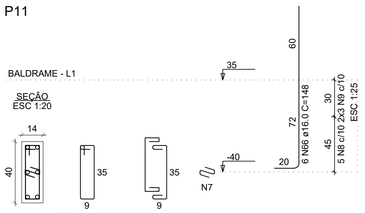
P5=P9



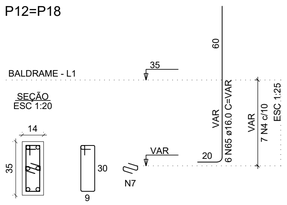
P1=P2



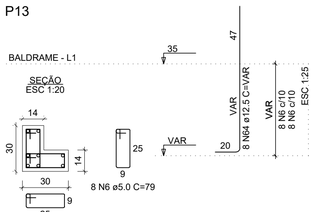
P11



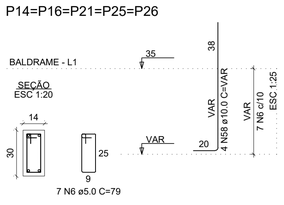
P12=P18



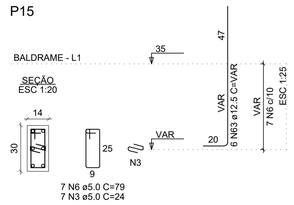
P13



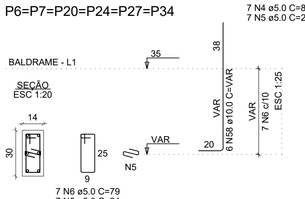
P14=P16=P21=P25=P26



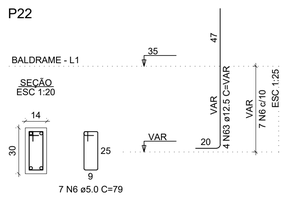
P15



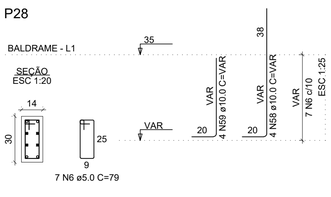
P6=P7=P20=P24=P27=P34



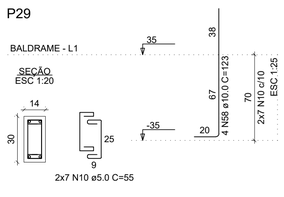
P22



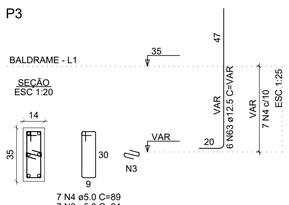
P28



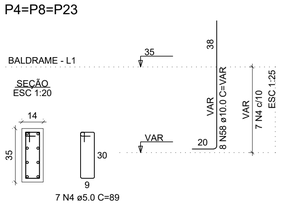
P29



P3



P4=P8=P23



PROJETO ESTRUTURAL

OBRA: CONSTRUÇÃO DE UMA EDIFICAÇÃO HABITACIONAL COM 2 PAVIMENTOS EM ALVENARIÁ

PROPR.:

PROJETO: Geminadas.com

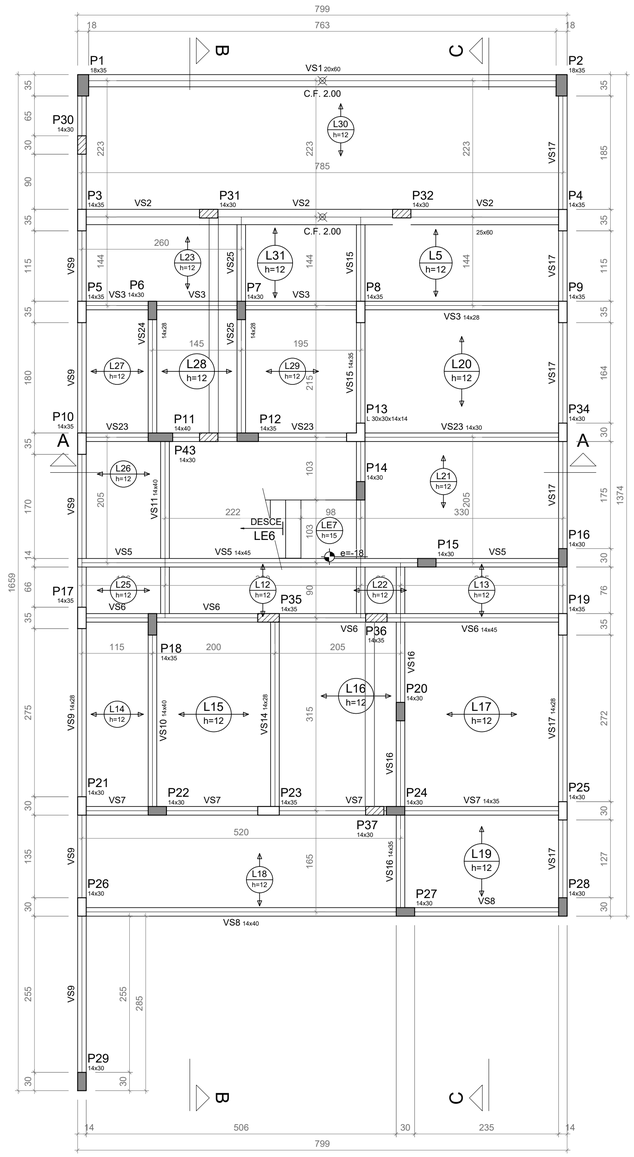
FOLHA Nº: 02/09

REVISÃO: R-01

REF.: DETALHAMENTO VIGAS E PILARES - PAV. BALDRAME

DATA: 03/2019

CONCRETO: 25 MPa

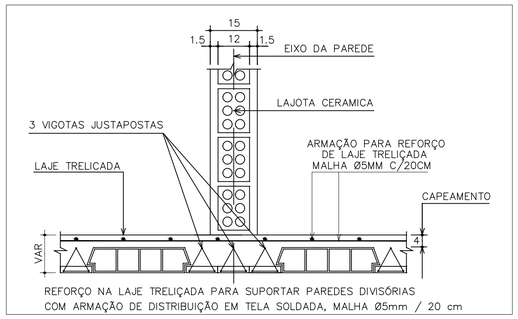


Forma do pavimento Superior (Nível 377)

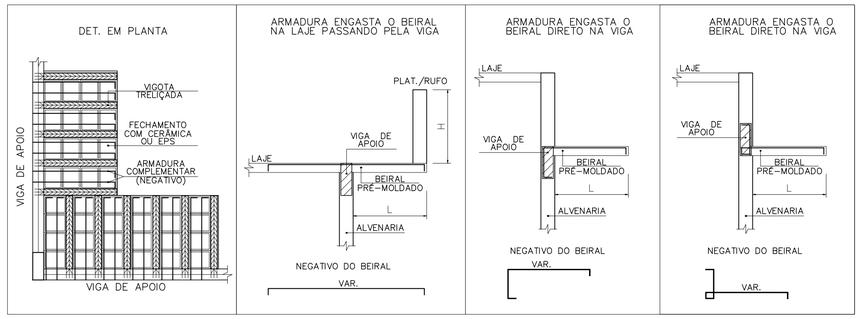
escala 1:50

Vigas			
Nome	Seção (cm)	Elevação (cm)	Nível (cm)
VS1	20x60	0	377
VS2	25x60	0	377
VS3	14x28	0	377
VS5	14x45	0	377
VS6	14x45	0	377
VS7	14x35	0	377
VS8	14x40	0	377
VS9	14x28	0	377
VS10	14x40	0	377
VS11	14x40	0	377
VS14	14x28	0	377
VS15	14x35	0	377
VS16	14x35	0	377
VS17	14x28	0	377
VS23	14x30	0	377
VS24	14x28	0	377
VS25	14x28	0	377

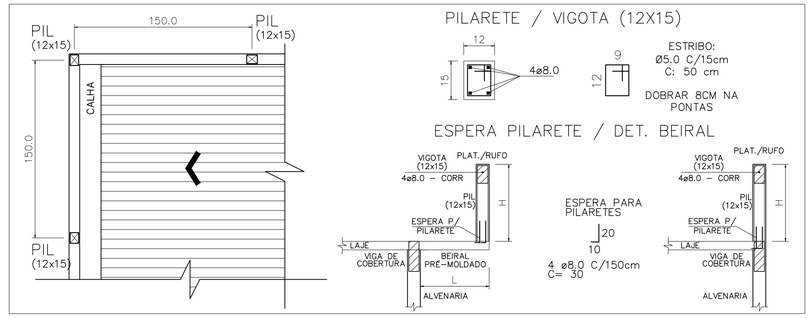
DETALHE GENÉRICO REFORÇO NA LAJE TRELIÇADA

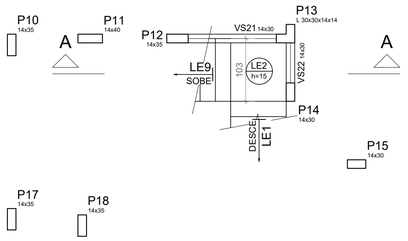


DETALHE GENÉRICO DE BEIRAL PRÉ MOLDADO



DETALHE GENÉRICO DE PLATIBANDA

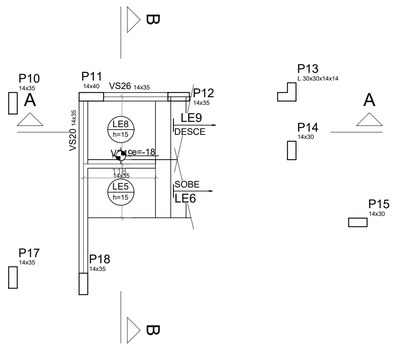




Forma intermediária do pavimento Superior (Nível 125)

escala 1:50

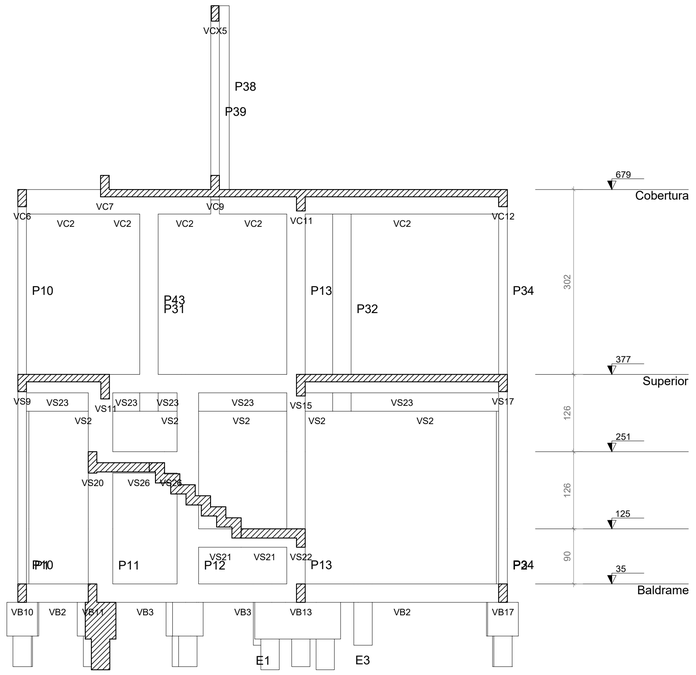
Vigas			
Nome	Seção (cm)	Elevação (cm)	Nível (cm)
VS21	14x30	0	125
VS22	14x30	0	125



Forma intermediária do pavimento Superior (Nível 251)

escala 1:50

Vigas			
Nome	Seção (cm)	Elevação (cm)	Nível (cm)
VS19	14x35	0	251
VS20	14x35	0	251
VS26	14x35	0	251



Corte A-A

escala 1:50

OBSERVAÇÕES		TABELA DE EQUIV. AÇOS	
SOBRECARGA NAS LAJES		mm	Pol.
		5,0	3/16"
		6,3	1/4"
200Kgf/m2	SEM ALVENARIA	8,0	5/16"
		10	3/8"
		12,5	1/2"
300Kgf/m2	COM ALVENARIA	16	5/8"
		20	3/4"

LEGENDA PILARES E VIGAS:

- Pilar que nasce
- Pilar que continua
- Pilar que morre
- Pilarete (PIL)
- Alvenaria sobre laje
- Pilar com mudança de seção
- Viga
- Viga chata ou invertida
- Lxx / hxx — Numero da laje, sentido da vigota e altura da laje

ESPECIFICAÇÕES:

– CONFRONTAR PROJETO ESTRUTURAL COM PROJETO ARQUITETÔNICO E LOCAÇÃO EM OBRA
– EM LOCAIS COM ATERRO, ACRESCENTAR A ALTURA DO ATERRO NA PROFUNDIDADE DAS FUNDAÇÕES E EM SUAS ARMADURAS.
– ANTES DO INÍCIO DA CONCRETAGEM, AS FORMAS DEVERÃO ESTAR LIMPAS E ESTANQUES DE MODO A EVITAR EVENTUAIS FUGAS DE PASTA.
– AS FORMA DEVERÃO SER MOLHADAS ATÉ A SATURAÇÃO A FIM DE EVITAR A ABSORÇÃO DA ÁGUA DE AMASSAMENTO DO CONCRETO
– AS BARRAS DE AÇO NÃO DEVEM APRESENTAR FERRUGEM, MANCHAS DE ÓLEO OU QUAISQUER OUTRAS SUBSTÂNCIAS QUE IMPEÇAM UMA PERFEITA ADERÊNCIA AO CONCRETO
– AS ARMADURAS NÃO DEVERÃO FICAR EM CONTATO DIRETO COM AS FORMAS, OBEDECENDO PARA ISSO OS COBRIMENTOS MÍNIMOS
– O AÇO UTILIZADO NA ESTRUTURA DEVE SER CA 50A E CA 60.
– O ADENSAMENTO É OBRIGATÓRIO E DEVERÁ SER CUIDADOSO, OCUPANDO TODOS OS RECANTOS DA FORMA, EVITANDO A VIBRAÇÃO DAS ARMADURAS QUE PODE PROVOCAR VAZIOS AO REDOR DAS ARMADURAS, DIFICULTANDO A ADERÊNCIA DO CONCRETO
– O PROCESSO DE CURA DO CONCRETO SERÁ NO MÍNIMO DE SETE DIAS

– PRAZOS RECOMENDADOS PARA DESFORMA:
  FACES LATERAIS: 03 DIAS
  FACES INFERIORES: 14 DIAS
  FACES INFERIORES SEM PONTALETES: 21 DIAS
– A RETIRADA DOS ESCORAMENTOS DOS TETOS DEVERÁ SER FEITA DE MANEIRA CONVENIENTE E PROGRESSIVA, PARTICULARMENTE PARA AS PEÇAS EM BALANÇO, O QUE IMPEDIRÁ O APARECIMENTO DE FISSURAS EM DECORRÊNCIA DE CARGAS DIFERENCIAIS
– A LAJE PRÉ-MOLDADA É DE RESPONSABILIDADE DA EMPRESA FORNECEDORA DE LAJES
– MEDIDAS APRESENTADAS NO PROJETO EM CENTÍMETROS (CM)
– QUALQUER DÚVIDA CONSULTAR O ENGENHEIRO CALCULISTA

CONCRETO: FCK=250Kgf/cm²

PROJETO ESTRUTURAL

OBRA: CONSTRUÇÃO DE UMA EDIFICAÇÃO HABITACIONAL COM 2 PAVIMENTOS EM ALVENARIA

PROPR.:

PROJETO: Geminadas.com

FOLHA Nº: 03/09

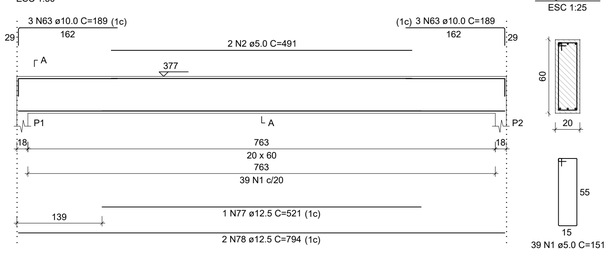
REVISÃO: R-01

DATA: 03/2019

REF.: FORMA - PAV. SUPERIOR E PAV. INTERMEDIARIOS (ESCADA)
CORTE AA
DETALHES GENÉRICOS

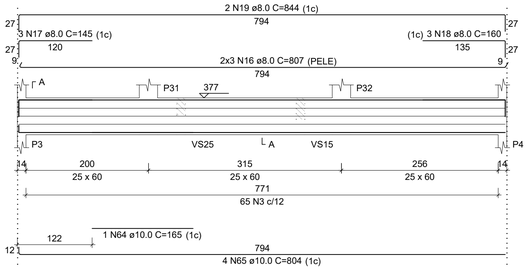
CONCRETO: 25 MPa

VS1
ESC 1:50

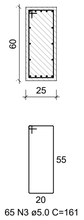


SEÇÃO A-A
ESC 1:25

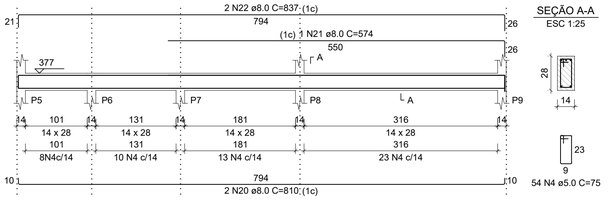
VS2
ESC 1:50



SEÇÃO A-A
ESC 1:25

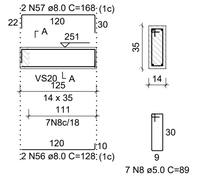


VS3
ESC 1:50



SEÇÃO A-A
ESC 1:25

VS19
ESC 1:50



SEÇÃO A-A
ESC 1:25

RELAÇÃO DO AÇO

VS1	VS2	VS3
VS5	VS6	VS7
VS8	VS9	VS10
VS11	VS14	VS15
VS16	VS17	VS19
VS20	VS21	

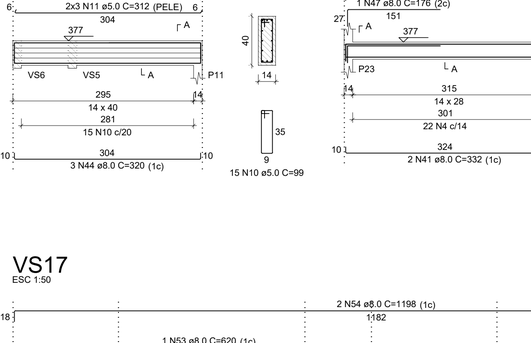
AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	39	151	5889
	2	5.0	2	491	982
	3	5.0	65	161	10465
	4	5.0	263	75	19725
	5	5.0	99	109	10791
	6	5.0	8	445	3560
	7	5.0	6	100	600
	8	5.0	162	88	14256
	9	5.0	5	24	120
	10	5.0	67	99	6633
	11	5.0	6	312	1872
	12	5.0	11	79	869
CA50	13	6.3	35	110	3850
	14	6.3	6	414	2484
	15	6.3	19	90	1710
	16	8.0	6	807	4842
	17	8.0	3	145	435
	18	8.0	3	160	480
	19	8.0	2	844	1688
	20	8.0	8	810	6480
	21	8.0	1	574	574
	22	8.0	2	837	1674
	23	8.0	1	426	426
	24	8.0	1	268	268
	25	8.0	1	195	195
	26	8.0	1	265	265
	27	8.0	1	707	707
	28	8.0	2	834	1668
	29	8.0	1	535	535
	30	8.0	2	380	760
	31	8.0	2	842	1684
	32	8.0	2	1084	2168
	33	8.0	3	265	795
	34	8.0	2	630	1260
	35	8.0	1	515	515
	36	8.0	1	270	270
	37	8.0	1	164	164
	38	8.0	1	167	167
	39	8.0	2	1198	2396
	40	8.0	2	556	1112
	41	8.0	4	332	1328
	42	8.0	1	179	179
	43	8.0	2	366	732
	44	8.0	3	320	960
	45	8.0	1	174	174
	46	8.0	2	352	704
	47	8.0	1	176	176
	48	8.0	1	180	180
	49	8.0	2	365	730
	50	8.0	2	685	1370
	51	8.0	2	335	670
	52	8.0	2	1164	2328
	53	8.0	1	620	620
	54	8.0	2	1198	2396
	55	8.0	2	261	522
	56	8.0	2	128	256
	57	8.0	2	168	336
	58	8.0	2	333	666
	59	8.0	2	345	690
	60	8.0	2	208	416
	61	8.0	3	214	642
	62	8.0	3	220	660
	63	10.0	6	189	1134
	64	10.0	1	165	165
	65	10.0	4	804	3216
	66	10.0	1	170	170
	67	10.0	2	425	850
	68	10.0	2	852	1704
	69	10.0	1	185	185
	70	10.0	4	411	1644
	71	10.0	2	811	1622
	72	10.0	8	101	808
	73	10.0	1	215	215
	74	10.0	2	405	810
	75	10.0	3	325	975
	76	10.0	2	612	1224
	77	12.5	1	521	521
	78	12.5	2	794	1588
	79	12.5	1	343	343
	80	12.5	1	347	347
	81	12.5	1	357	357
	82	12.5	2	846	1692
	83	12.5	2	389	778
	84	12.5	2	397	794
	85	12.5	1	250	250
	86	12.5	2	253	506
	87	12.5	2	257	514
	88	12.5	1	201	201
	89	12.5	1	251	251
	90	12.5	2	258	516

RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	6.3	80.4	21.7
	8.0	482.6	209.5
	10.0	147.2	99.8
	12.5	86.6	91.7
CA60	5.0	759.2	128.7
PESO TOTAL (kg)			
CA50	422.7		
CA60	128.7		

Volume de concreto (C-25) = 6.92 m³
Área de forma = 104.67 m²

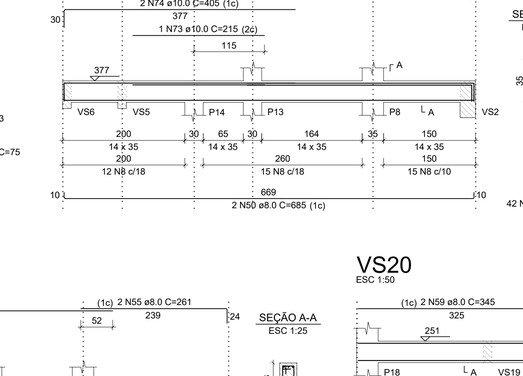
VS5
ESC 1:50



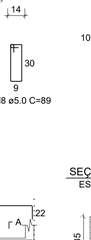
SEÇÃO A-A
ESC 1:25



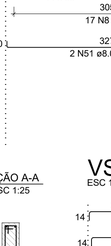
VS6
ESC 1:50



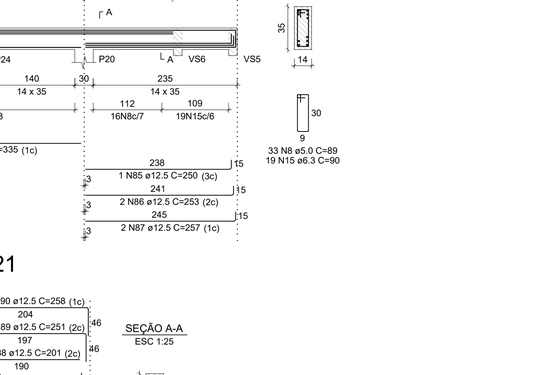
SEÇÃO A-A
ESC 1:25



ESPERA P35
ESC 1:25



VS7
ESC 1:50



SEÇÃO A-A
ESC 1:25



VS8
ESC 1:50

SEÇÃO A-A
ESC 1:25

VS9
ESC 1:50

SEÇÃO A-A
ESC 1:25

VS10
ESC 1:50

SEÇÃO A-A
ESC 1:25

VS11
ESC 1:50

SEÇÃO A-A
ESC 1:25

VS14
ESC 1:50

SEÇÃO A-A
ESC 1:25

VS15
ESC 1:50

SEÇÃO A-A
ESC 1:25

VS16
ESC 1:50

SEÇÃO A-A
ESC 1:25

VS17
ESC 1:50

SEÇÃO A-A
ESC 1:25

VS20
ESC 1:50

SEÇÃO A-A
ESC 1:25

VS21
ESC 1:50

SEÇÃO A-A
ESC 1:25

PROJETO ESTRUTURAL

OBRA: CONSTRUÇÃO DE UMA EDIFICAÇÃO HABITACIONAL COM 2 PAVIMENTOS EM ALVENARIA

PROPR.:

PROJETO: Geminadas.com

FOLHA Nº: 04/09

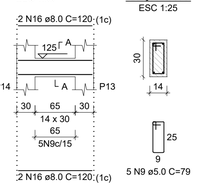
REVISÃO: R-01

DATA: 03/2019

REF.: DETALHAMENTO VIGAS E PILARES (01/03) - PAV. SUPERIOR

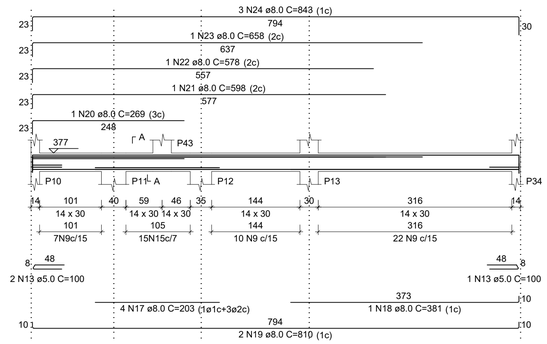
CONCRETO: 25 MPa

VS22
ESC 1:50

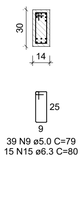


SEÇÃO A-A
ESC 1:25

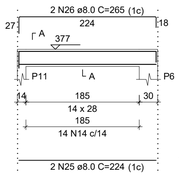
VS23
ESC 1:50



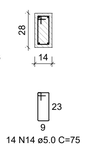
SEÇÃO A-A
ESC 1:25



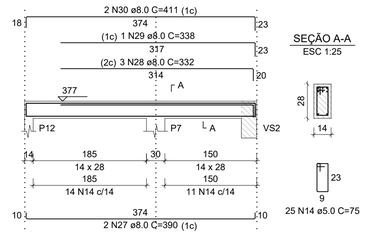
VS24
ESC 1:50



SEÇÃO A-A
ESC 1:25

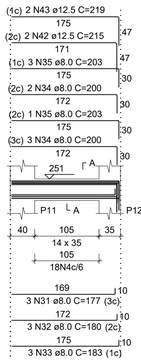


VS25
ESC 1:50

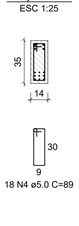


SEÇÃO A-A
ESC 1:25

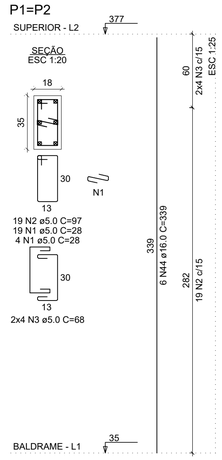
VS26
ESC 1:50



SEÇÃO A-A
ESC 1:25



P1=P2



RELAÇÃO DO AÇO

2xP1	P10	P11
P12	P13	P14
P15	P16	P17
P18	P19	2xP20
P21	P22	P25
P26	VS22	VS23
VS24	VS25	VS26

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	46	28	1288
	2	5.0	38	97	3686
	3	5.0	16	68	1088
	4	5.0	172	89	15308
	5	5.0	42	60	2520
	6	5.0	105	24	2520
	7	5.0	31	99	3069
	8	5.0	8	65	520
	9	5.0	391	79	30889
	10	5.0	76	55	4180
	11	5.0	35	24	840
	12	5.0	70	24	1680
	13	5.0	3	100	300
	14	5.0	39	75	2925
CA50	15	6.3	15	80	1200
	16	8.0	4	120	480
	17	8.0	4	203	812
	18	8.0	1	381	381
	19	8.0	2	810	1620
	20	8.0	1	269	269
	21	8.0	1	598	598
	22	8.0	1	578	578
	23	8.0	1	658	658
	24	8.0	3	843	2529
	25	8.0	2	224	448
	26	8.0	2	265	530
	27	8.0	2	390	780
	28	8.0	3	332	996
	29	8.0	1	338	338
	30	8.0	2	411	822
	31	8.0	3	177	531
	32	8.0	3	180	540
	33	8.0	3	183	549
	34	8.0	5	200	1000
	35	8.0	4	203	812
	36	10.0	10	75	750
	37	10.0	24	380	9120
	38	10.0	20	339	6780
	39	12.5	8	94	752
	40	12.5	8	389	3112
	41	12.5	10	339	3390
	42	12.5	2	215	430
	43	12.5	2	219	438
	44	16.0	30	339	10170

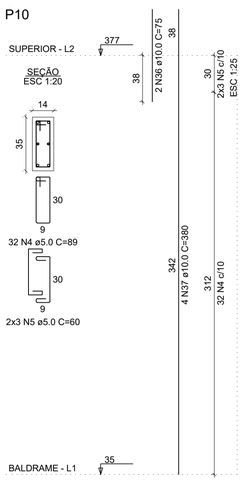
RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	6.3	12	3.2
	8.0	152.7	66.3
	10.0	166.5	112.9
	12.5	81.2	86.1
	16.0	101.7	176.6
CA60	5.0	708.1	120.1

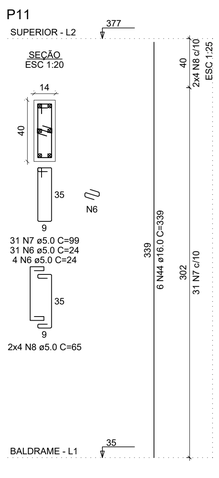
PESO TOTAL (kg)	
CA50	445.1
CA60	120.1

Volume de concreto (C-25) = 3.69 m³
Área de forma = 71.50 m²

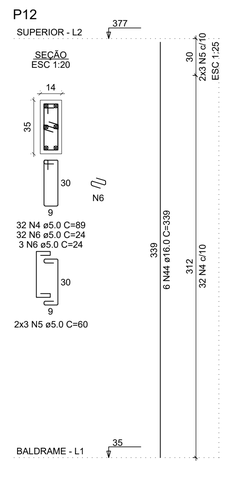
P10



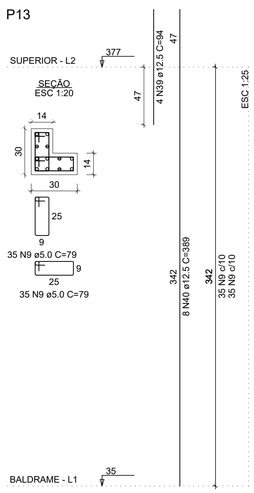
P11



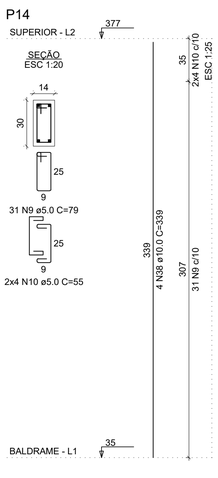
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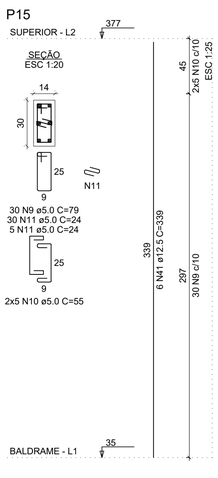
P13



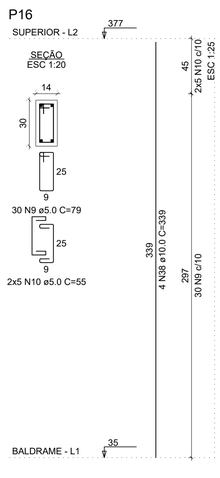
P14



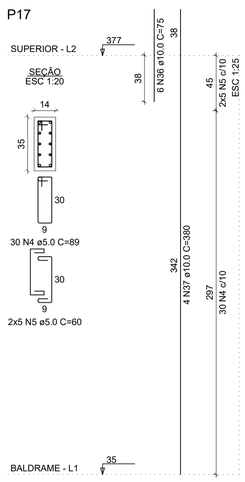
P15



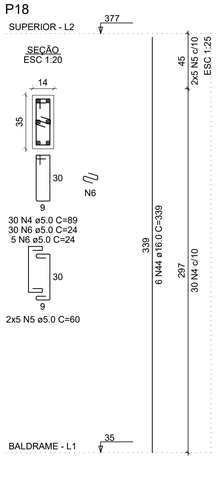
P16



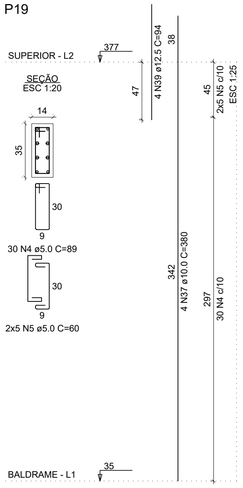
P17



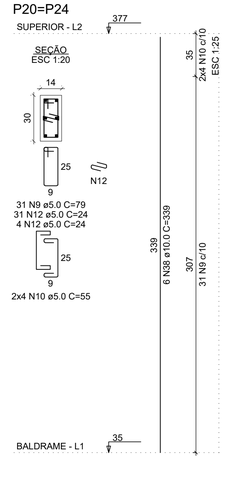
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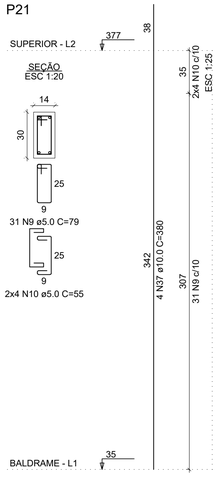
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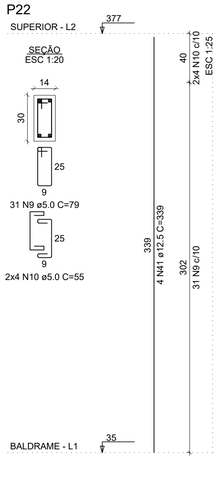
P20=P24



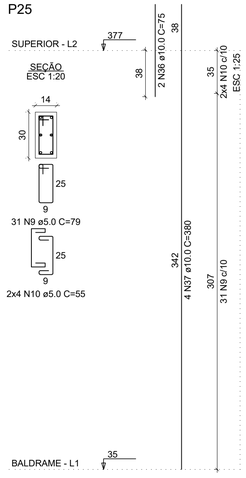
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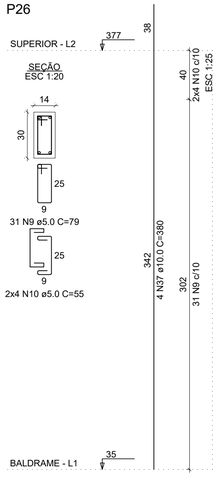
P22



P25



P26



PROJETO ESTRUTURAL

OBRA: CONSTRUÇÃO DE UMA EDIFICAÇÃO HABITACIONAL COM 2 PAVIMENTOS EM ALVENARIA

PROPR.:

PROJETO: Geminadas.com

FOLHA Nº: 05/09

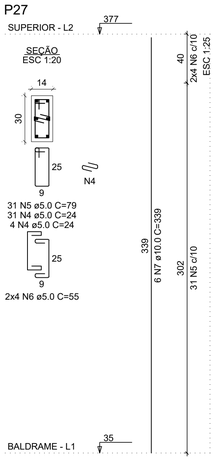
REVISÃO: R-01

DATA: 03/2019

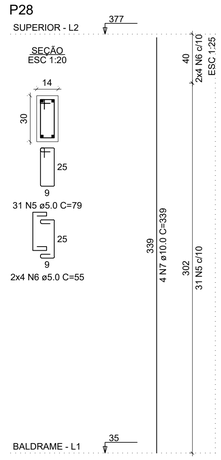
REF.: CONT. DETALHAMENTO VIGAS E PILARES (02/03) - PAV. SUPERIOR

CONCRETO: 25 MPa

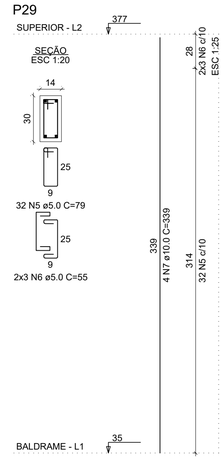
P27



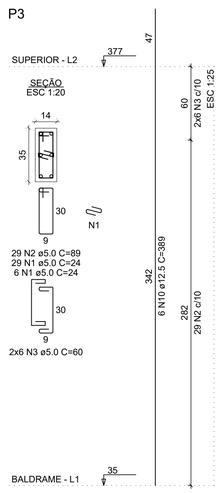
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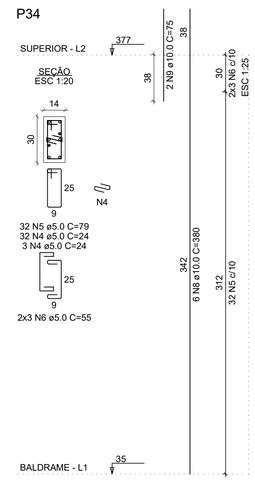
P29



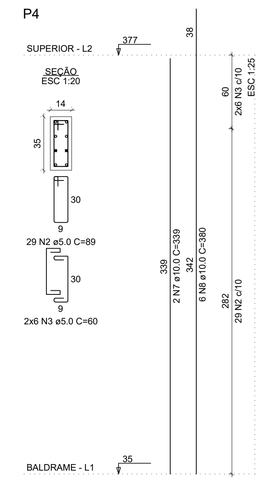
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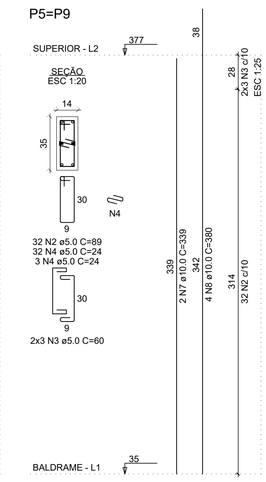
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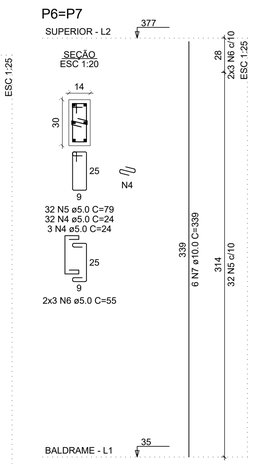
P4



P5=P9



P6=P7



RELAÇÃO DO AÇO

P3 P4 2xP5
2xP6 2xP8 P27
P28 P29 P34

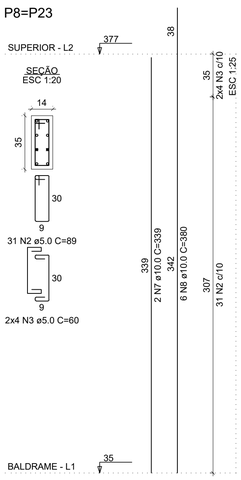
AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	35	24	840
	2	5.0	184	89	16376
	3	5.0	52	60	3120
	4	5.0	210	24	5040
	5	5.0	190	79	15010
	6	5.0	40	55	2200
CA50	7	10.0	36	339	12204
	8	10.0	32	380	12160
	9	10.0	2	75	150
	10	12.5	6	389	2334

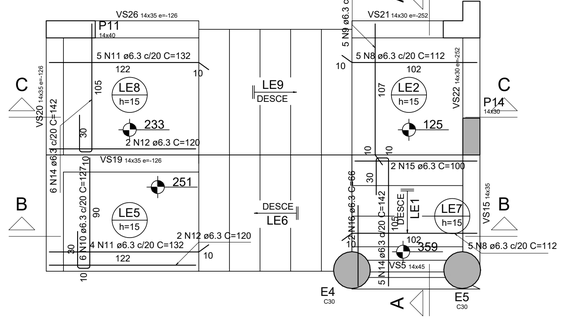
RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	10.0	245.1	166.3
	12.5	23.3	24.7
CA60	5.0	425.9	72.2
PESO TOTAL (kg)			
CA50	191		
CA60	72.2		

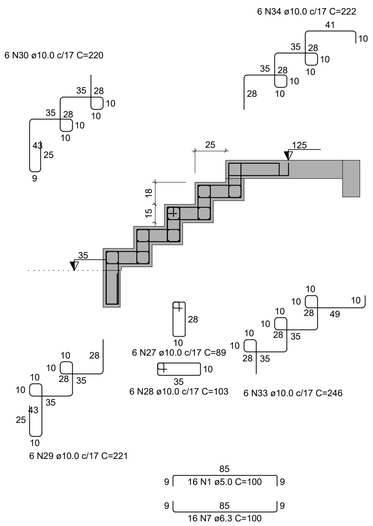
Volume de concreto (C-25) = 1.87 m³
Área de forma = 38.17 m²

P8=P23

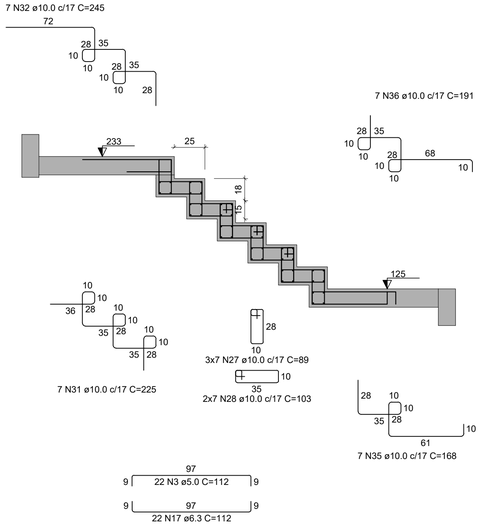




Armação positiva da escada E1
ESC 1:25



Corte A-A (LE1)
ESC 1:25



Corte C-C (LE9)
ESC 1:25

RELAÇÃO DO AÇO

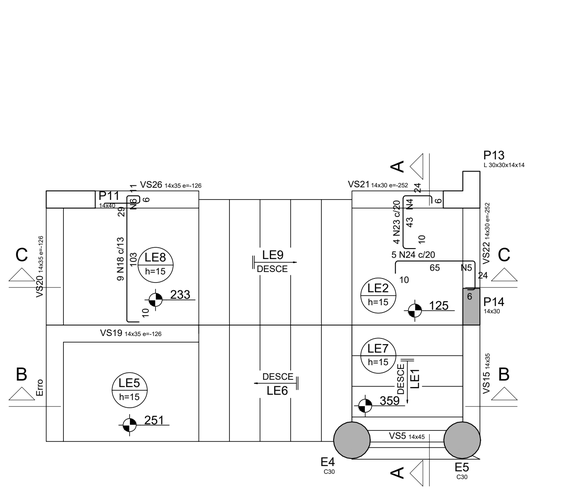
LE1 LE2 LE5
LE6 LE7 LE8
LE9 Negativos

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	16	100	1600
	2	5.0	22	105	2310
	3	5.0	22	112	2464
	4	5.0	3	84	252
	5	5.0	4	105	420
	6	5.0	6	118	708
CA50	7	6.3	16	100	1600
	8	6.3	10	112	1120
	9	6.3	5	117	585
	10	6.3	6	127	762
	11	6.3	9	132	1188
	12	6.3	4	120	480
	13	6.3	22	104	2288
	14	6.3	11	142	1562
	15	6.3	2	100	200
	16	6.3	2	66	132
	17	6.3	22	112	2464
	18	6.3	9	153	1377
	19	8.0	21	88	1848
	20	8.0	14	102	1428
	21	8.0	7	181	1267
	22	8.0	7	234	1638
	23	8.0	4	78	312
	24	8.0	5	100	500
	25	8.0	7	300	2100
	26	8.0	7	259	1813
	27	10.0	27	89	2403
	28	10.0	14	103	1442
	29	10.0	6	221	1326
	30	10.0	6	220	1320
	31	10.0	7	225	1575
	32	10.0	7	245	1715
	33	10.0	6	246	1476
	34	10.0	6	222	1332
	35	10.0	7	168	1176
	36	10.0	7	191	1337

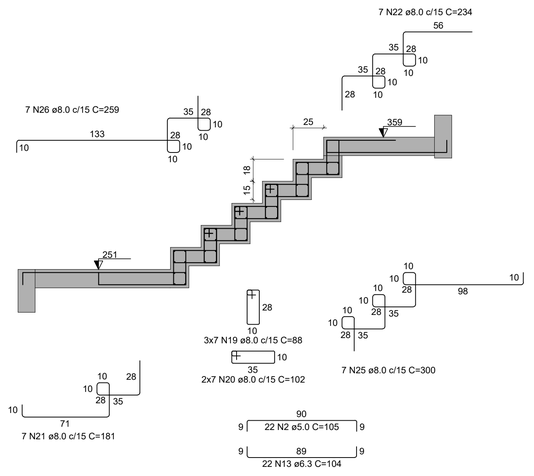
RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	6.3	137.6	37
	8.0	109.1	47.3
	10.0	151	102.4
CA60	5.0	77.5	13.1
PESO TOTAL (kg)			
CA50	186.8		
CA60	13.1		

Volume de concreto (C-25) = 1.48 m³
Área de forma = 16.08 m²



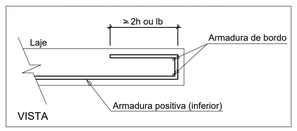
Armação negativa da escada E1
ESC 1:25



Corte B-B (LE6)
ESC 1:25

Armaduras de distribuição	
Armadura	Armadura de distribuição
N23	3 N4 ø5.0 c/17 C=84
N24	4 N5 ø5.0 c/17 C=105
N18	6 N6 ø5.0 c/17 C=118

DETALHE DA ARMADURA
DE BORDO LIVRE DA LAJE



PROJETO ESTRUTURAL

OBRA: CONSTRUÇÃO DE UMA EDIFICAÇÃO HABITACIONAL COM 2 PAVIMENTOS EM ALVENARIA

PROPR.:

PROJETO: Geminadas.com

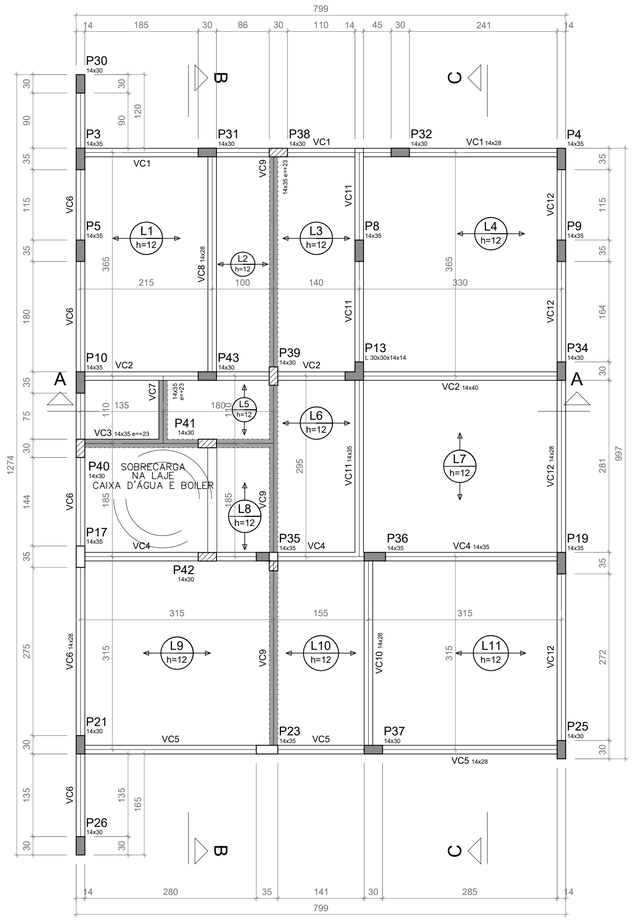
FOLHA Nº: 06/09

REVISÃO: R-01

DATA: 03/2019

REF.: CONT. DETALHAMENTO VIGAS E PILARES (03/03) - PAV. SUPERIOR
DETALHAMENTO ESCADAS

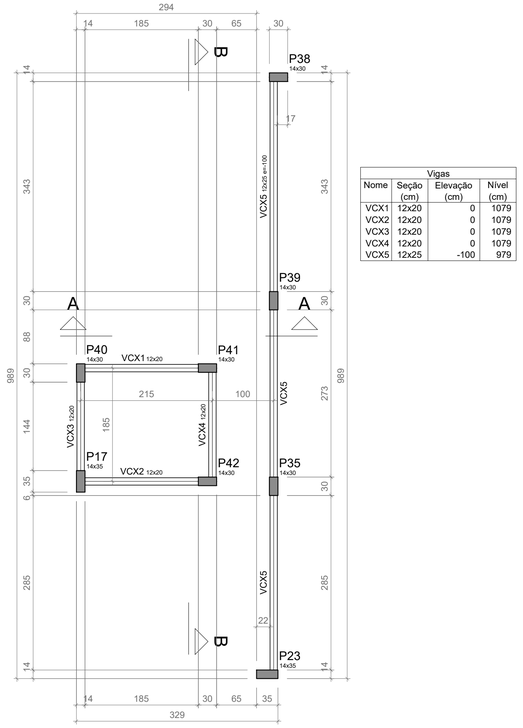
CONCRETO: 25 MPa



Forma do pavimento Cobertura (Nível 679)

escala 1:50

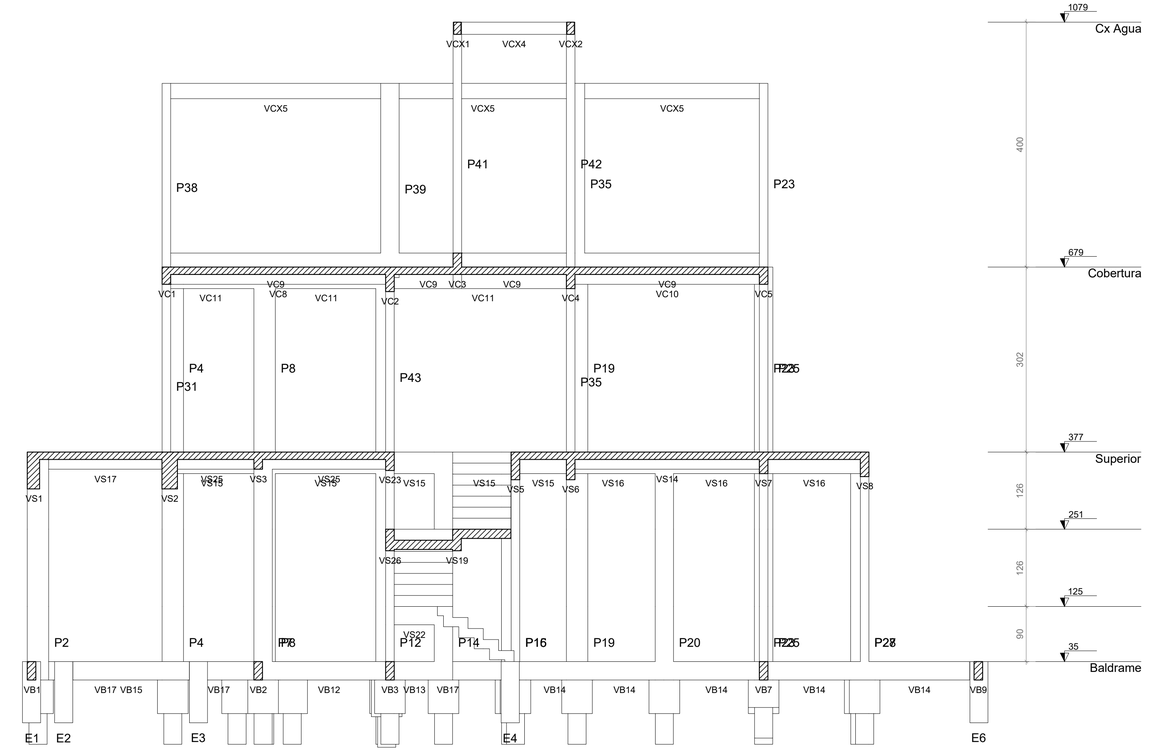
Vigas			
Nome	Seção (cm)	Elevação (cm)	Nível (cm)
VC1	14x28	0	679
VC2	14x40	0	679
VC3	14x35	23	702
VC4	14x35	0	679
VC5	14x28	0	679
VC6	14x28	0	679
VC7	14x35	23	702
VC8	14x28	0	679
VC9	14x35	23	702
VC10	14x28	0	679
VC11	14x35	0	679
VC12	14x28	0	679



Vigas			
Nome	Seção (cm)	Elevação (cm)	Nível (cm)
VCX1	12x20	0	1079
VCX2	12x20	0	1079
VCX3	12x20	0	1079
VCX4	12x20	0	1079
VCX5	12x25	-100	979

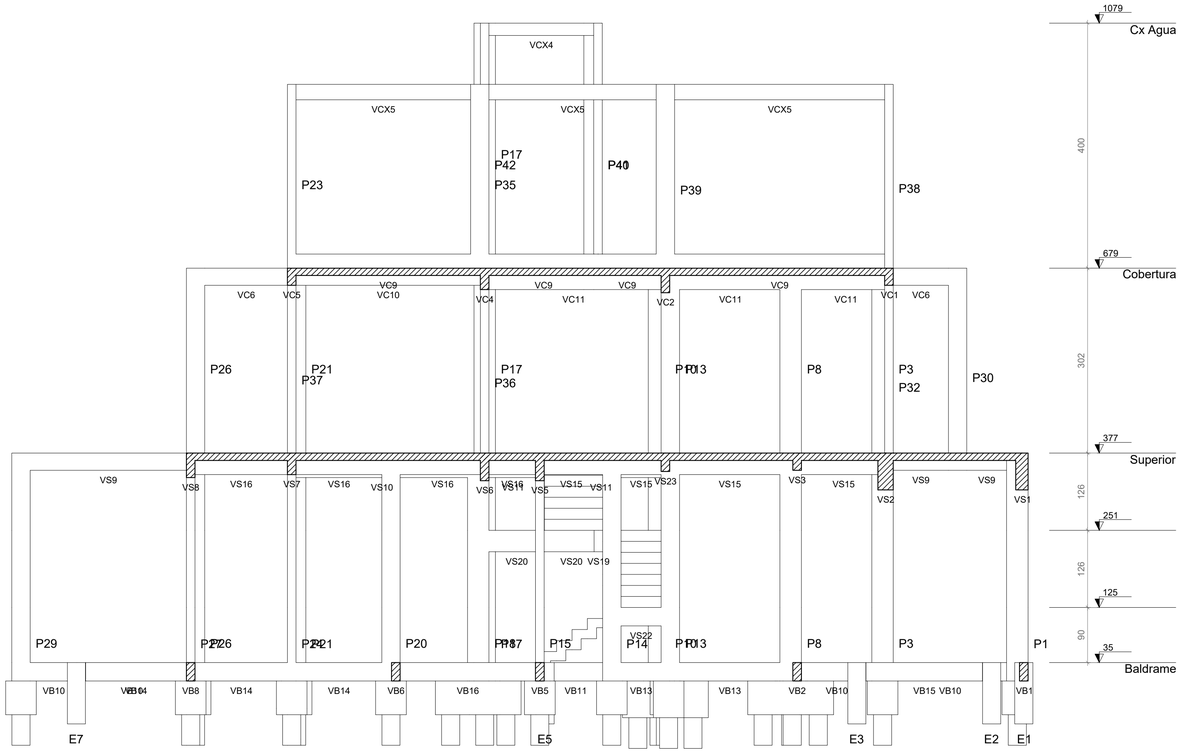
Forma do pavimento Cx Agua (Nível 1079)

escala 1:50



Corte B-B

escala 1:50



Corte C-C

escala 1:50

OBSERVAÇÕES		TABELA DE EQUIV. AÇOS	
SOBRECARGA NAS LAJES		mm	Pol.
		5,0	3/16"
		6,3	1/4"
200Kgf/m2	SEM ALVENARIA	8,0	5/16"
		10	3/8"
		12,5	1/2"
300Kgf/m2	COM ALVENARIA	16	5/8"
		20	3/4"

LEGENDA PILARES E VIGAS:

- Pilar que nasce
- Pilar com mudança de seção
- Pilar que continua
- Viga
- Pilar que morre
- Viga chata ou Invertida
- Pilarete (PIL)
- Numero da laje, sentido da vigota e altura da laje (Lxx Vxx)
- Alvenaria sobre laje

ESPECIFICAÇÕES:

– CONFRONTAR PROJETO ESTRUTURAL COM PROJETO ARQUITETÔNICO E LOCAÇÃO EM OBRA
– EM LOCAIS COM ATERRO, ACRESCENTAR A ALTURA DO ATERRO NA PROFUNDIDADE DAS FUNDAÇÕES E EM SUAS ARMADURAS.
– ANTES DO INÍCIO DA CONCRETAGEM, AS FORMAS DEVERÃO ESTAR LIMPAS E ESTANQUES DE MODO A EVITAR EVENTUAIS FUGAS DE PASTA.
– AS FORMA DEVERÃO SER MOLHADAS ATÉ A SATURAÇÃO A FIM DE EVITAR A ABSORÇÃO DA ÁGUA DE AMASSAMENTO DO CONCRETO
– AS BARRAS DE AÇO NÃO DEVEM APRESENTAR FERRUGEM, MANCHAS DE ÓLEO OU QUAISQUER OUTRAS SUBSTÂNCIAS QUE IMPEÇAM UMA PERFEITA ADERÊNCIA AO CONCRETO
– AS ARMADURAS NÃO DEVERÃO FICAR EM CONTATO DIRETO COM AS FORMAS, OBEDECENDO PARA ISSO OS COBRIMENTOS MÍNIMOS
– O AÇO UTILIZADO NA ESTRUTURA DEVE SER CA 50A E CA 60.
– O ADENSAMENTO É OBRIGATÓRIO E DEVERÁ SER CUIDADOSO, OCUPANDO TODOS OS RECANTOS DA FORMA, EVITANDO A VIBRAÇÃO DAS ARMADURAS QUE PODE PROVOCAR VAZIOS AO REDOR DAS ARMADURAS, DIFICULTANDO A ADERÊNCIA DO CONCRETO
– O PROCESSO DE CURA DO CONCRETO SERÁ NO MÍNIMO DE SETE DIAS

– PRAZOS RECOMENDADOS PARA DESFORMA:
  FACES LATERAIS: 03 DIAS
  FACES INFERIORES: 14 DIAS
  FACES INFERIORES SEM PONTALETES: 21 DIAS
– A RETIRADA DOS ESCORAMENTOS DOS TETOS DEVERÁ SER FEITA DE MANEIRA CONVENIENTE E PROGRESSIVA, PARTICULARMENTE PARA AS PEÇAS EM BALANÇO, O QUE IMPEDIRÁ O APARECIMENTO DE FISSURAS EM DECORRÊNCIA DE CARGAS DIFERENCIAIS
– A LAJE PRÉ–MOLDADA É DE RESPONSABILIDADE DA EMPRESA FORNECEDORA DE LAJES
– MEDIDAS APRESENTADAS NO PROJETO EM CENTÍMETROS (CM)
– QUALQUER DÚVIDA CONSULTAR O ENGENHEIRO CALCULISTA

CONCRETO: FCK=250Kgf/cm²

PROJETO ESTRUTURAL

OBRA: CONSTRUÇÃO DE UMA EDIFICAÇÃO HABITACIONAL COM 2 PAVIMENTOS EM ALVENARIA
RUA OCTÁVIO MARCHI , N°277, JD ESPANHA, LT 11, QD 125
MARINGÁ - PARANÁ.

PROPR.:

PROJETO: Geminadas.com

FOLHA Nº: 07/09

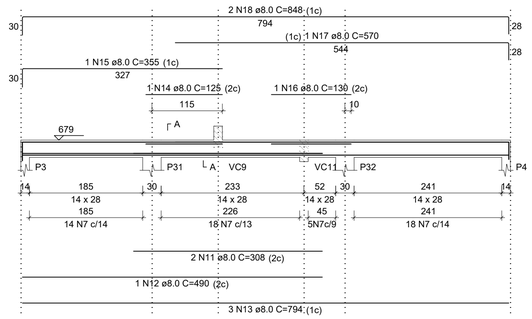
REVISÃO: R-01

DATA: 03/2019

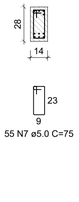
REF.: FORMA - PAV. COBERTURA
FORMA - PAV. CX ÁGUA
CORTES BB e CC

CONCRETO: 25 MPa

VC1
ESC 1:50

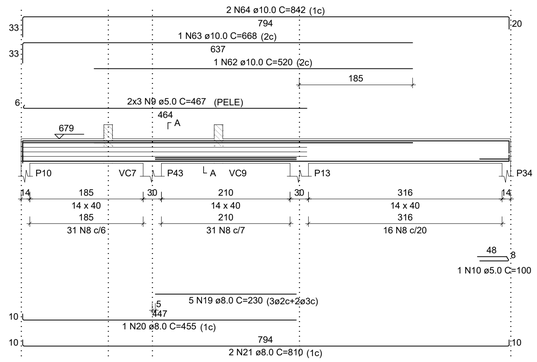


SEÇÃO A-A
ESC 1:25

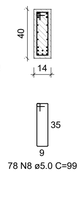


55 N7 ø5.0 C=75

VC2
ESC 1:50

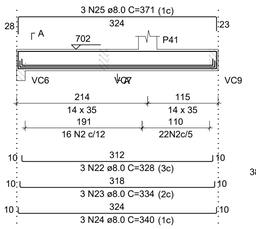


SEÇÃO A-A
ESC 1:25

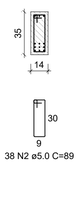


78 N8 ø5.0 C=99

VC3
ESC 1:50

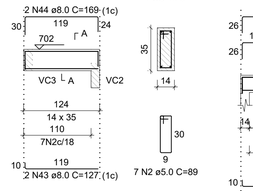


SEÇÃO A-A
ESC 1:25



38 N2 ø5.0 C=89

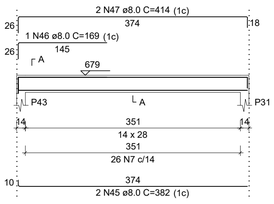
VC7
ESC 1:50



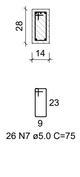
SEÇÃO A-A
ESC 1:25

7 N2 ø5.0 C=89

VC8
ESC 1:50



SEÇÃO A-A
ESC 1:25



26 N7 ø5.0 C=75

RELAÇÃO DO AÇO

P10	P13	P17
P19	2xP21	P23
VC1	VC2	VC3
VC4	VC5	VC6
VC7	VC8	VC9
VC10	VC11	VC12

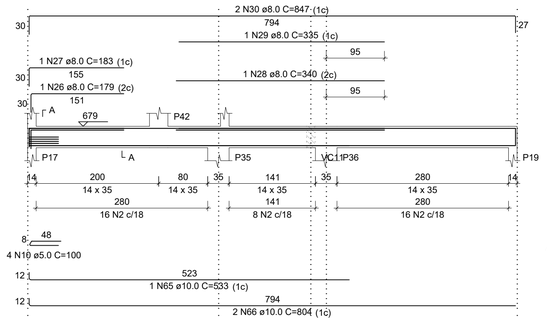
AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	62	24	1488
	2	5.0	292	89	25988
	3	5.0	30	60	1800
	4	5.0	118	79	9322
	5	5.0	12	55	660
	6	5.0	31	24	744
	7	5.0	294	75	22050
	8	5.0	78	99	7722
	9	5.0	6	467	2802
	10	5.0	5	100	500
CA50	11	8.0	2	308	616
	12	8.0	1	490	490
	13	8.0	3	794	2382
	14	8.0	1	125	125
	15	8.0	1	355	355
	16	8.0	1	130	130
	17	8.0	1	570	570
	18	8.0	2	848	1696
	19	8.0	5	230	1150
	20	8.0	1	455	455
	21	8.0	4	810	3240
	22	8.0	3	328	984
	23	8.0	3	334	1002
	24	8.0	6	340	2040
	25	8.0	3	371	1113
	26	8.0	1	179	179
	27	8.0	1	183	183
	28	8.0	1	340	340
	29	8.0	1	335	335
	30	8.0	2	847	1694
	31	8.0	2	234	468
	32	8.0	1	379	379
	33	8.0	1	499	499
	34	8.0	2	826	1652
	35	8.0	3	270	810
	36	8.0	2	1164	2328
	37	8.0	2	165	330
	38	8.0	1	290	290
	39	8.0	1	580	580
	40	8.0	1	926	926
	41	8.0	2	1166	2332
	42	8.0	2	203	406
	43	8.0	2	127	254
	44	8.0	2	169	338
	45	8.0	2	382	764
	46	8.0	1	169	169
	47	8.0	2	414	828
	48	8.0	1	626	626
	49	8.0	2	1000	2000
	50	8.0	1	173	173
	51	8.0	1	180	180
	52	8.0	6	390	2340
	53	8.0	2	1032	2064
	54	8.0	2	365	730
	55	8.0	2	685	1370
	56	8.0	2	992	1984
	57	8.0	1	285	285
	58	8.0	2	1032	2064
	59	10.0	16	299	4784
	60	10.0	8	340	2720
	61	10.0	8	75	600
	62	10.0	1	520	520
	63	10.0	1	668	668
	64	10.0	2	842	1684
	65	10.0	1	533	533
	66	10.0	2	804	1608
	67	10.0	2	220	440
	68	10.0	1	470	470
	69	10.0	2	712	1424
	70	12.5	20	299	5980
	71	12.5	6	349	2094

RESUMO DO AÇO

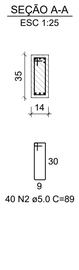
AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	8.0	462.5	200.7
	10.0	154.5	104.8
	12.5	80.7	85.6
CA60	5.0	730.8	123.9
PESO TOTAL (kg)			
CA50	391.1		
CA60	123.9		

Volume de concreto (C-25) = 4.71 m³
Área de forma = 84.81 m²

VC4
ESC 1:50

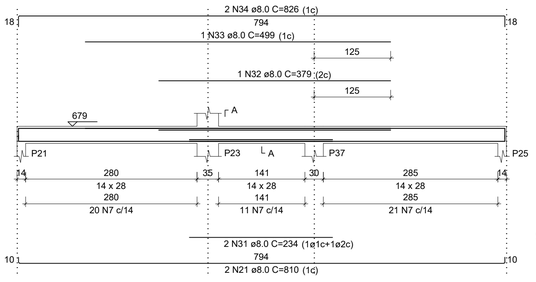


SEÇÃO A-A
ESC 1:25

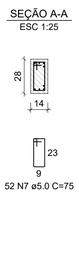


40 N2 ø5.0 C=89

VC5
ESC 1:50

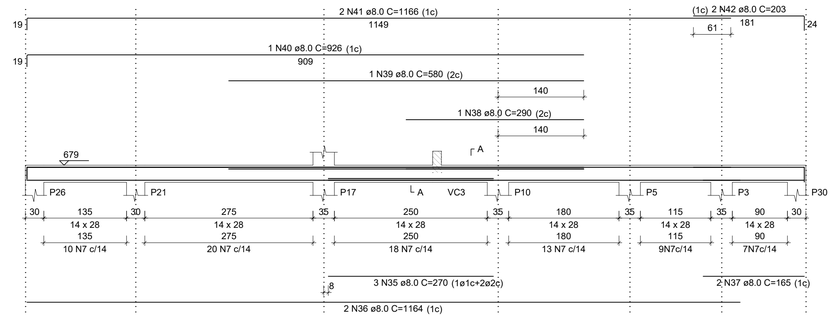


SEÇÃO A-A
ESC 1:25

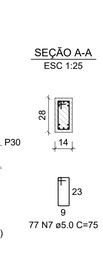


52 N7 ø5.0 C=75

VC6
ESC 1:50

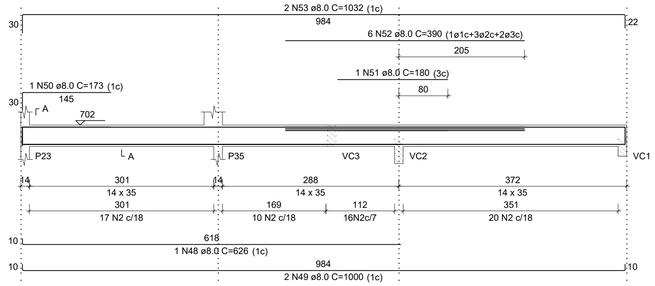


SEÇÃO A-A
ESC 1:25

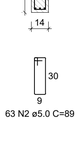


77 N7 ø5.0 C=75

VC9
ESC 1:50

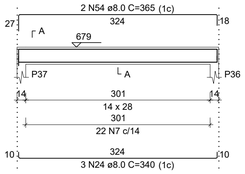


SEÇÃO A-A
ESC 1:25

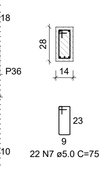


63 N2 ø5.0 C=89

VC10
ESC 1:50

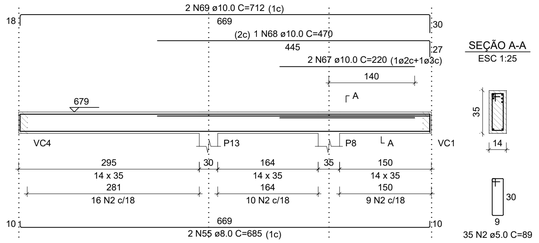


SEÇÃO A-A
ESC 1:25

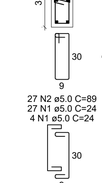


22 N7 ø5.0 C=75

VC11
ESC 1:50

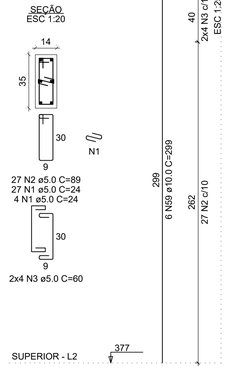


SEÇÃO A-A
ESC 1:25

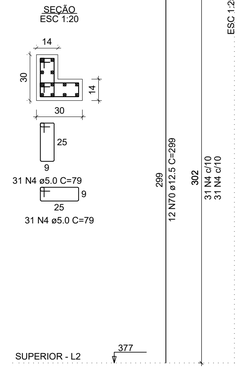


35 N2 ø5.0 C=89

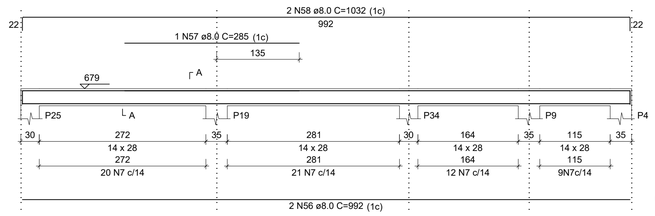
P10



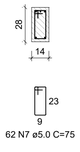
P13



VC12
ESC 1:50

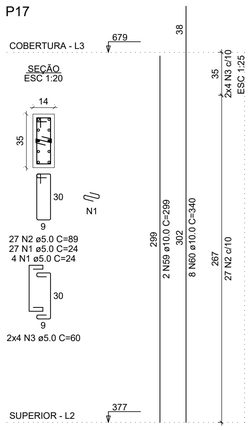


SEÇÃO A-A
ESC 1:25

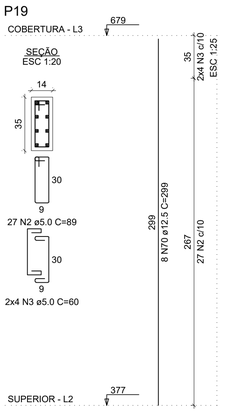


62 N7 ø5.0 C=75

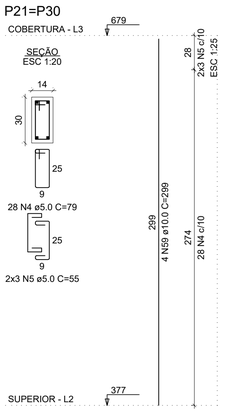
P17



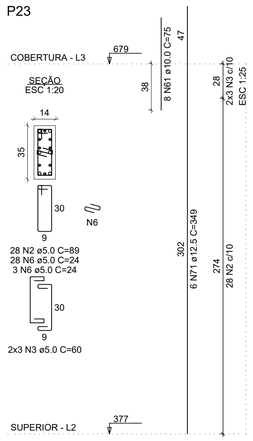
P19



P21=P30



P23



PROJETO ESTRUTURAL

OBRA: CONSTRUÇÃO DE UMA EDIFICAÇÃO HABITACIONAL COM 2 PAVIMENTOS EM ALVENARIA

PROPR.:

PROJETO: Geminadas.com

FOLHA Nº: 08/09

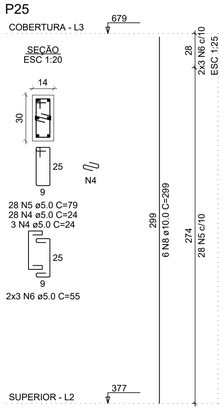
REVISÃO: R-01

DATA: 03/2019

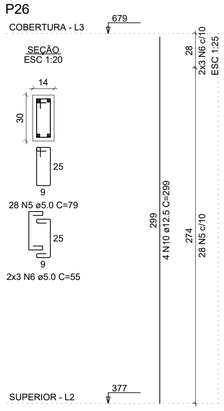
REF.: DETALHAMENTO VIGAS E PILARES - PAV. COBERTURA

CONCRETO: 25 MPa

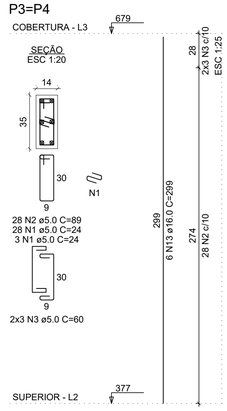
P25



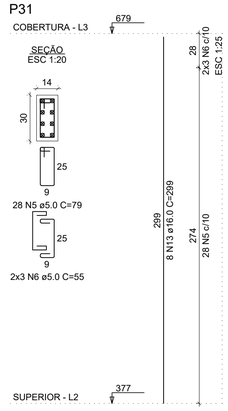
P26



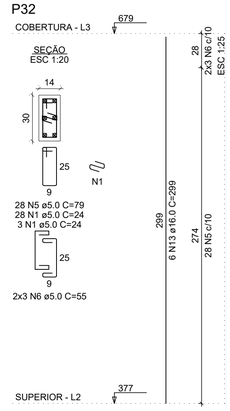
P3=P4



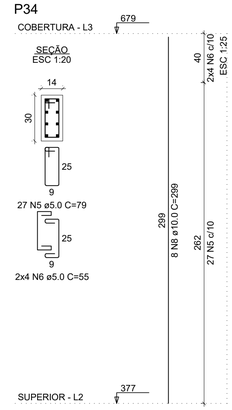
P31



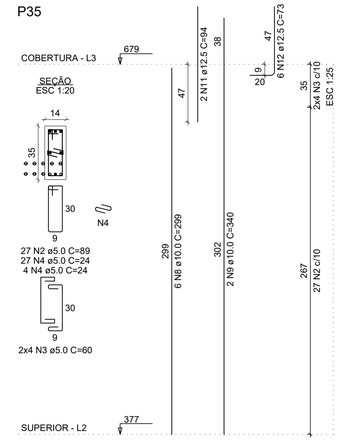
P32



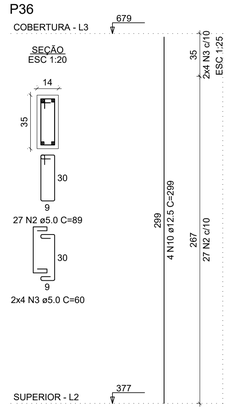
P34



P35



P36



RELAÇÃO DO AÇO

2xP3	2xP5	P8
P25	P26	P31
P32	P34	P35
P36	P37	P43

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	93	24	2232
	2	5.0	193	89	17177
	3	5.0	48	60	2880
	4	5.0	93	24	2232
	5	5.0	194	79	15326
	6	5.0	46	55	2530
	7	5.0	31	24	744
CA50	8	10.0	34	299	10166
	9	10.0	2	340	680
	10	12.5	14	299	4186
	11	12.5	2	94	188
	12	12.5	6	73	438
	13	16.0	30	299	8970

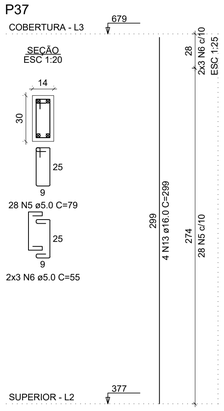
RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	10.0	108.5	73.6
	12.5	48.1	51
	16.0	89.7	155.7
CA60	5.0	431.2	73.1

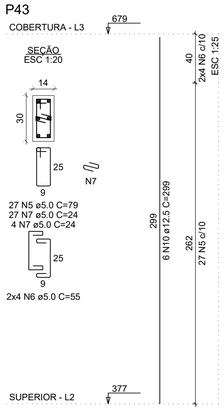
PESO TOTAL (kg)	
CA50	280.3
CA60	73.1

Volume de concreto (C-25) = 1.92 m³
Área de forma = 39.32 m²

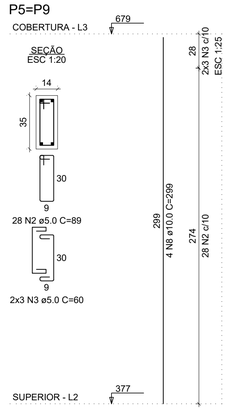
P37



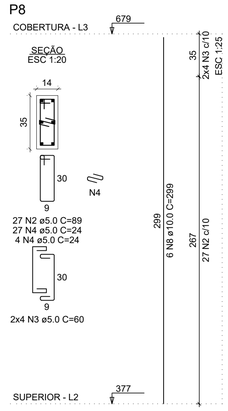
P43



P5=P9

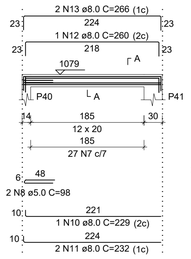


P8

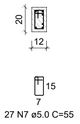


VCX1

ESC 1:50

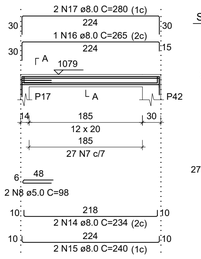


SEÇÃO A-A
ESC 1:25

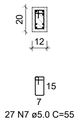


VCX2

ESC 1:50

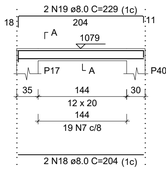


SEÇÃO A-A
ESC 1:25

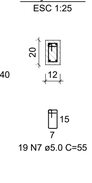


VCX3

ESC 1:50

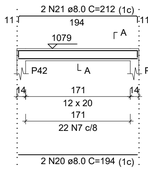


SEÇÃO A-A
ESC 1:25

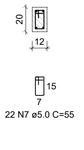


VCX4

ESC 1:50

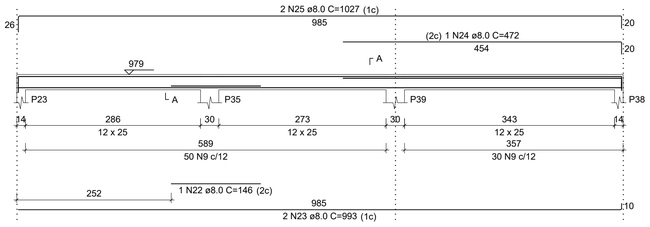


SEÇÃO A-A
ESC 1:25

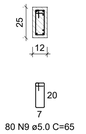


VCX5

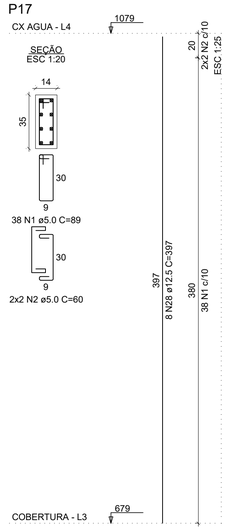
ESC 1:50



SEÇÃO A-A
ESC 1:25



P17



RELAÇÃO DO AÇO

P17	P23	P35
P38	P39	P40
P41	P42	VCX1
VCX2	VCX3	VCX4
VCX5		

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	66	89	5874
	2	5.0	10	60	600
	3	5.0	102	24	2448
	4	5.0	129	24	3096
	5	5.0	194	79	15326
	6	5.0	30	55	1650
	7	5.0	95	55	5225
	8	5.0	4	98	392
	9	5.0	80	65	5200
CA50	10	8.0	1	229	229
	11	8.0	2	232	464
	12	8.0	1	260	260
	13	8.0	2	266	532
	14	8.0	2	234	468
	15	8.0	2	240	480
	16	8.0	1	265	265
	17	8.0	2	280	560
	18	8.0	2	204	408
	19	8.0	2	229	458
	20	8.0	2	194	388
	21	8.0	2	212	424
	22	8.0	1	146	146
	23	8.0	2	993	1986
	24	8.0	1	472	472
	25	8.0	2	1027	2054
	26	10.0	14	297	4158
	27	10.0	8	397	3176
	28	12.5	22	397	8734
	29	12.5	10	297	2970
	30	12.5	16	274	4384

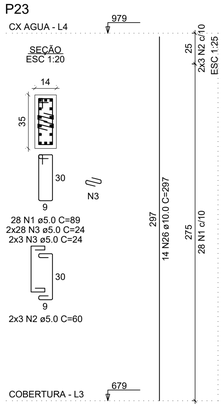
RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	8.0	95.9	41.6
	10.0	73.3	49.7
	12.5	160.9	170.5
CA60	5.0	398.1	67.5

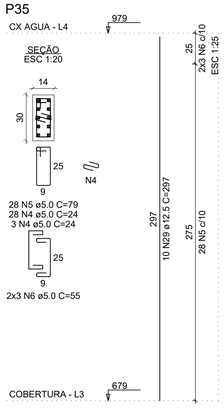
PESO TOTAL (kg)	
CA50	261.9
CA60	67.5

Volume de concreto (C-25) = 1.71 m³
Área de forma = 35.58 m²

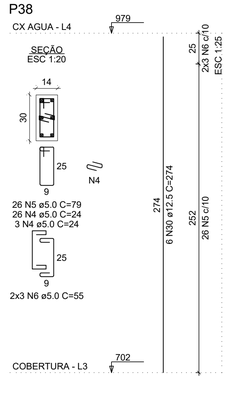
P23



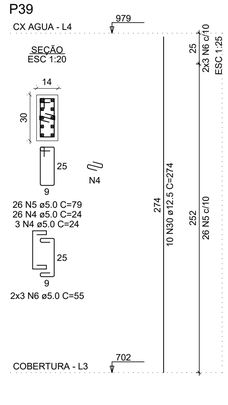
P35



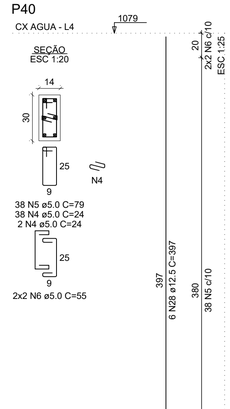
P38



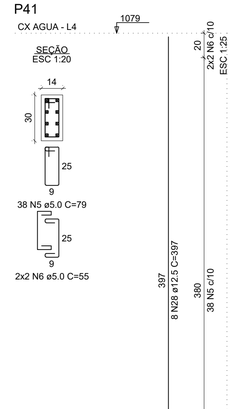
P39



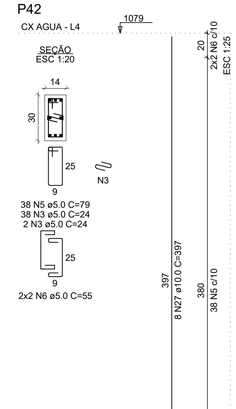
P40



P41



P42



PROJETO ESTRUTURAL

OBRA: CONSTRUÇÃO DE UMA EDIFICAÇÃO HABITACIONAL COM 2 PAVIMENTOS EM ALVENARIA

PROPR.:

PROJETO:

Geminadas.com

FOLHA Nº: 09/09

REVISÃO: R-01

DATA: 03/2019

REF.: CONT. DETALHAMENTO VIGAS E PILARES - PAV. COBERTURA
DETALHAMENTO VIGAS E PILARES - PAV. CX ÁGUA

CONCRETO: 25 MPa